



Contents lists available at ScienceDirect

Journal of Financial Economics

journal homepage: www.elsevier.com/locate/jfecConvertible bond arbitrage, liquidity externalities, and stock prices[☆]Darwin Choi^{a,1}, Mila Getmansky^{b,2}, Heather Tookes^{a,*}^a Yale School of Management, P.O. Box 208000, New Haven, CT 06520-8000, USA^b Isenberg School of Management, University of Massachusetts, 121 Presidents Drive, Room 308C, Amherst, MA 01003, USA

ARTICLE INFO

Article history:

Received 19 March 2007

Received in revised form

23 January 2008

Accepted 25 February 2008

Available online 25 November 2008

JEL classification:

G12

G14

Keywords:

Convertible bond arbitrage

Liquidity

Market efficiency

Hedge funds

ABSTRACT

In the context of convertible bond issuance, we examine the impact of arbitrage activity on underlying equity markets. In particular, we use changes in equity short interest following convertible bond issuance to identify convertible bond arbitrage activity and analyze its impact on stock market liquidity and prices for the period 1993 to 2006. There is considerable evidence of arbitrage-induced short selling resulting from issuance. Moreover, we find strong evidence that this activity is systematically related to liquidity improvements in the stock. These results are robust to controlling for the potential endogeneity of arbitrage activity.

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1. Introduction

Does arbitrage activity impact market quality? Although this question is not new, the proliferation of

hedge funds in recent years has brought increasing attention to important questions regarding their impact on both liquidity and market efficiency (see, e.g., [Securities and Exchange Commission \(SEC\) Staff Report, 2003](#)). In this paper, we focus on one particular strategy: convertible bond arbitrage. The growth in the issuance of the equity-linked debt securities can be attributed, at least in part, to the growing supply of capital provided by hedging strategies. Convertible bond issuance has increased more than sixfold in the past 15 years, from \$7.8 billion in 1992 to \$50.2 billion in 2006 (Securities Data Corporation (SDC), Global New Issues database). In fact, the widespread belief among Wall Street practitioners is that convertible bond arbitrage hedge funds purchase 70% to 80% of the convertible debt offered in primary markets.¹

[☆] We would like to thank Vikas Agarwal, Nick Bollen, Ben Branch, John Burke, Michael Epstein, Richard Evans, Shuang Feng, Laura Frieder, William Fung, Paul Gao, William Goetzmann, Robin Greenwood, Jennifer Jeurgens, Charles Jones, Nikunj Kapadia, Hossein Kazemi, Camelia Kuhnen, Owen Lamont, Laura Lindsay, David Modest, Sanjay Nawalkha, Thomas Schneeweis, Matthew Spiegel, Norman Wechsler, Rebecca Zarutskie, and seminar participants at Yale University, the Batten Young Finance Scholars Conference, UMASS-Amherst, the NBER Microstructure working group, and the FDIC/JFSR Bank Research Conference on Liquidity and Liquidity Risk for helpful discussions. We are extremely grateful to Paul Bennett and the NYSE for providing the short-interest data. We also thank Eric So for his assistance with the Nasdaq data and Scott Zhu for excellent research assistance. We are grateful to an anonymous referee for suggestions that greatly improved the paper. We thank the International Center of Finance at the Yale School of Management for its financial support. Any errors are our own.

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¹ While they do not constitute the entire universe of convertible bond arbitrageurs, hedge funds are an important subset. [Mitchell, Pedersen, and Pulvino \(2007\)](#), report that convertible arbitrage funds account for 75% of the market. Similar estimates can be found in the popular press. See, e.g., a *Wall Street Journal* article ([Pulliam, 2004](#)) on convertible bond issuance in 2004: "As much as 80% of those issues were bought by hedge funds, according to brokers who work on

In order to clarify the intuition as to why convertible bond arbitrage might impact liquidity in underlying equity markets, it is useful to outline the basics of the strategy. The aim of convertible bond arbitrage is to exploit mispricing in convertible bonds, usually by buying an undervalued convertible bond (Henderson, 2005) and taking a short position in the equity.² A typical convertible bond arbitrage strategy employs delta-neutral hedging, in which an arbitrageur buys the convertible bond and sells short the underlying equity at the current delta. There are two important components of the strategy (shown in Fig. 1). The first is the initial position, which is set up so that no profit or loss is generated from very small movements in the underlying stock price and where cash flows are captured from both the convertible bond's yield and the short position's interest rebate. The second is dynamic hedging activity that follows the initial short position. If the price of the stock increases, the arbitrageur adds to the short position because the delta has increased. Similarly, when the stock price declines, the arbitrageur buys stock to cover part of the short position due to the decrease in delta. Aggregate equity market trading demand, in contrast, is expected to move in the opposite direction. For example, Chordia, Roll, and Subrahmanyam (2002) show a positive correlation between stock returns and order imbalances. This means that the dynamic hedging activities of convertible bond arbitrageurs, a class of investors trading against net market demand, should improve liquidity. This potentially positive role for hedge funds and other convertible bond arbitrageurs is contrary to the view of a destabilizing role for arbitrageurs in markets (see Mayhew, 2000, for a survey of this literature).

Although we do not have direct data on convertible bond arbitrage activity in individual stocks, we are able to identify firms and dates on which we know that initial arbitrage positions are taken: convertible bond issuance dates. We use these initial positions as a proxy for the presence of convertible bond arbitrageurs in the market for the stock (i.e., their future dynamic hedging). For the period 1993 to 2006, we calculate changes in short interest at issuance. Our approach is simple, yet it captures the strategy, as we observe large increases in short interest near convertible debt offerings. The methodology allows us to use aggregate data to identify the presence of a particular type of trader in equity markets.

Our proxy for arbitrage activity (initial change in short interest of issuing firms) has several advantages over using hedge fund databases to estimate convertible bond

arbitrage activity. First, this provides a measure of positions taken by arbitrageurs in individual securities. Fund flows data in hedge fund databases are self-reported and therefore provide an incomplete measure of convertible bond arbitrage activity. The databases only partially represent the hedge fund universe, with many large funds choosing not to participate. Second, there can be style misclassification and funds reporting multiple strategies to hedge fund databases. Third, even if we measured the assets of the funds perfectly, the positions would still be unobservable due to the use of leverage.

We find considerable evidence of arbitrage activity (i.e., short selling in the stock at the date of bond issuance). We also find increased equity market liquidity following bond issuance. Moreover, these liquidity improvements are positively and significantly related to our proxy for convertible bond arbitrage activity. We also observe changes in stock return volatility. Following convertible debt issuance there is an average decrease in total return volatility as well as the idiosyncratic component of volatility. However, we do not find evidence of a systematic relationship between convertible bond arbitrage activity and these changes. We measure price efficiency using return autocorrelation and variance ratios (as in Lo and MacKinlay, 1988), to capture the extent to which stock prices follow a random walk. We do not observe significant changes in either of these measures following issuance. Taken together, we interpret the findings as evidence that convertible bond arbitrage activity tends to positively affect equity markets; however, this is primarily through liquidity improvements, not through stock prices.

A critical aspect of the analysis is that we do not observe arbitrage activity directly. Instead, we infer it based on changes in short interest at bond issuance. We conduct several tests to examine the validity of this important assumption.³ First, we rule out the possibility that changes that we observe are due to changes in market-wide variables or to factors impacting firms with similar characteristics. We do this by conducting all analyses based on changes relative to a set of control firms (matched on industry, exchange, size, book-to-market, and turnover). Second, it could be that the short selling that we observe is due to valuation shorting resulting from news of the convertible bond issue, not due to classic convertible bond arbitrage. In order to address this issue, we hand-collect announcement dates for our sample of issues. The announcement and issue dates allow us to separate the impact of announcement period shorting versus issue period shorting, which we interpret as valuation shorting versus convertible bond arbitrage, respectively. In all of these tests, we find evidence consistent with the view that the short selling that we observe near convertible bond issues is due to convertible bond arbitrage. We also conduct robustness tests, in which we explicitly control for other potential sources of volume that can be associated with the

(footnote continued)

convertible-bond trading desks." *The Financial Times* (Skorecki, 2004) reports that hedge funds bought 70% of new issues in 2003 and that 95% of trades in converts are made by hedge funds. The evidence presented in this study of large increases in short selling near issuance is consistent with that view.

² A convertible bond is a hybrid debt instrument: it is a bond that may, at the option of the holder, be converted into stock at a specified price for a given time period. Due to the conversion option, convertible bond purchasers may profit from equity price gains, but they also have downside protection since they are guaranteed bond payments (and, in the event of bankruptcy, are senior to equity holders).

³ We thank an anonymous referee for encouraging this line of inquiry.

convertible bond issue. In addition, we control for potential endogeneity of arbitrage activity and find similar results.

The main contribution of this paper is that we identify arbitrage activity and are able to estimate its impact on market quality (we use changes in equity market liquidity and price efficiency as measures of quality). By identifying a particular trader type, our methodology allows us to shed additional light on the mechanisms through which quality changes following issuance occur. The primary findings suggest that changes in liquidity vary systematically with the positions taken by arbitrageurs. The findings in this paper may be of interest to managers of issuing firms concerned about liquidity and efficiency spillovers in their stock as a result of their capital structure decisions.

This paper is organized as follows. Section 2 contains a brief review of related literature. Section 3 constructs the main hypotheses. Section 4 describes the data and sample. Section 5 presents the analysis of arbitrage activity, liquidity, and prices. Several robustness checks are also presented. Finally, Section 6 concludes.

2. Related literature

The notions of liquidity and efficiency “externalities” underlie much of the analysis in this paper. The idea in Ross (1976) and subsequent theoretical works (e.g., Easley, O’Hara, and Srinivas, 1998; Biais and Hillion, 1994; Grossman, 1988) that the introduction of options markets can enhance efficiency by making markets less incomplete or by positively impacting informativeness of stock prices has been followed by empirical investigations of the impact of derivatives markets on the market for the underlying asset (e.g., Kumar, Sarin, and Shastri, 1998; DeTemple and Jorion, 1990).⁴ Mayhew (2000) provides an excellent survey of this literature. The main findings indicate that derivatives markets have a positive impact on liquidity and no negative impact on price efficiency. Most authors report a decrease in total volatility and an increase in trading volume following the introduction of options. We consider our study of the liquidity and efficiency externalities of convertible bond markets to be an extension of this line of research. Because of the embedded option in the convertible bond, the issuance of convertible bonds is analogous to the introduction of options.⁵ Our identification (based on short selling) allows us to provide a more direct test of the impact of arbitrageurs. While prior work has provided evidence that new securities markets can impact equity market quality on average, we identify the mechanisms through which quality changes occur.

Our basic empirical strategy uses increases in short interest near debt issuance to identify arbitrage activity. In that way, it is closely related to the growing empirical literature on short selling activity. There has been considerable focus on the relationship between future stock returns and both observed short sales and short-sales constraints (see, e.g., Diether, Lee, and Werner, 2008; Boehme, Danielsen, and Sorescu, 2006; Asquith, Pathak, and Ritter, 2005; Jones and Lamont, 2002; Dechow, Hutton, Meulbroek, and Sloan, 2001; Asquith and Meulbroek, 1996). The information content of short sales in event settings has also received attention in the recent empirical literature (e.g., Christophe, Ferri, and Angel, 2004). All of these papers provide evidence that short selling and short-sales constraints impact stock prices, suggesting that short sellers help to incorporate negative information into prices.

Although short sellers can help facilitate the incorporation of negative information into prices, many are uninformed. They use short sales to hedge other positions. Little has been done to distinguish this type of short seller.⁶ This is an important distinction because the impact of short selling on market quality will obviously depend largely on who is engaging in the short sale. Uninformed short sellers are likely to add liquidity to markets (rather than reduce it as a result of potential adverse selection). Asquith, Pathak, and Ritter (2005, p. 270) note that, “Of course, a firm might have a high short-interest ratio because there is both valuation shorting, and arbitrage shorting taking place simultaneously. Unfortunately, we cannot identify these situations precisely.” Our event-based approach takes us further towards identifying this specialized investment strategy from the aggregate data and distinguishing this activity from valuation shorting.

Three recent papers use changes in short interest near events to infer the impact of a particular type of trader. Arnold et al. (2005) use the Tax Payer Relief Act of 1997, which made selling short against the box more costly, as a laboratory for testing hypotheses regarding changes in the information content of short interest when tax-motivated short sellers (i.e., uninformed sellers) no longer have incentives to short. This event-driven approach to trader identification is similar in spirit to ours; however, we examine not only average changes, but also cross-sectional implications of the introduction of a particular trader type. That is, we examine the sensitivity of subsequent changes in liquidity and efficiency to the magnitude of the increase in short selling due to arbitrage. Mitchell, Pulvino, and Stafford (2004) use short interest in acquirers near merger announcements to identify activities by risk arbitrageurs and estimate their impact on prices. Bechmann (2004) provides evidence that short selling induced by hedging activities explains part of the stock price decline following convertible bond calls.

⁴ More recently, Basak and Croitoru (2006) show how the presence of arbitrageurs improves market quality and risk sharing in the context of rational markets with heterogeneous risk-averse investors and short-sales constraints.

⁵ In fact, in unreported analysis, we find that the absence of put or call options on a particular stock is associated with greater convertible bond arbitrage activity. This confirms the idea that the existence of substitute markets is critical in any trading decision.

⁶ Boehmer, Jones, and Zhang (2008) use proprietary order-level data from the NYSE to quantify the information content of the flow of shorting activity by the type of account initiating the sale. Their focus is on characterizing the information content of short sales, by size and trader (account type).

In both [Bechmann \(2004\)](#) and [Mitchell, Pulvino, and Stafford \(2004\)](#), the focus is mainly on price pressure induced by short selling activity while our focus is on the impact of arbitrage on stock market liquidity and prices.

Although they do not constitute the entire universe of convertible bond arbitrageurs, convertible bond arbitrage hedge funds do play a role in primary issues of convertible debt and can impact stock market quality. [Henderson \(2005\)](#) studies the underpricing of convertible bonds at issue, as well as the risk and returns of the convertible bond arbitrage strategy.⁷ He finds that new issues of convertible bonds are underpriced at issue but that excess returns occur soon after issuance (mainly in the first six months). This can decrease the presence of convertible bond arbitrageurs over longer horizons.⁸ [Mitchell, Pedersen, and Pulvino \(2007\)](#) analyze the impact of capital outflows in hedge funds on convertible bond prices. Finally, [Choi, Getmansky, Henderson, and Tookes \(2008\)](#) examine supply and demand in the convertible bond market. They map the measure of arbitrage activity used in this paper to fund flows and returns in convertible bond arbitrage hedge funds. Overall, findings in these papers suggest a significant role for hedge funds in convertible bond and related markets.

3. Arbitrage, liquidity, and stock prices: predictions

This section outlines the main predictions. We measure changes in short interest near convertible bond issuance and relate this to changes in liquidity and stock price efficiency. We test the following two null hypotheses:

H₀ (Liquidity): Convertible bond arbitrage activity, proxied by increased short interest near issuance, is uncorrelated with changes in liquidity.

H₀ (Efficiency): Convertible bond arbitrage activity is uncorrelated with changes in efficiency.

The typical convertible bond arbitrage strategy (delta hedging) implies that arbitrageurs engaged in dynamic hedging are likely to trade in the opposite direction of the rest of the market: they increase their short positions as stock prices increase, and decrease them when stock prices decrease. This should result in improved market liquidity (the alternative hypothesis).

The expected improvement in liquidity described above assumes that convertible bond arbitrageurs have no special knowledge about the value of the underlying shares. If they are instead privately informed about future stock values, adverse selection costs can increase, and equity market liquidity can decrease. While it is an empirical question, we do not expect to observe evidence of this because convertible bond arbitrageurs typically act to exploit perceived underpricing in the bond, not equity.

In addition to liquidity changes, convertible bond arbitrageurs can also impact the efficiency of equity prices. In theory, if the short selling that we identify in

the data is due to an informational advantage about equity market valuation, price efficiency would increase following issuance.⁹ Even if these short sellers are not privately informed but are trading to exploit a known inefficiency such as autocorrelation, efficiency will also increase following issuance.¹⁰ On the other hand, if short sellers are taking equity market positions primarily to hedge their positions in the bonds, then their presence would not directly impact efficiency of stock prices. We conjecture that although convertible bond arbitrageurs are sophisticated traders, they are relatively uninformed. That is, they have no private information about the value of the equity that they short. They are trading to manage equity risk exposure, not to exploit mispricing.¹¹ If this is the case, we predict:

P1: Convertible bond arbitrage activity, proxied by the increase in short interest near issuance, will be associated with improved market liquidity. This occurs via dynamic hedging strategies, in which arbitrageurs' trading activity tends to be in the opposite direction of the market.

P2: Convertible bond arbitrage activity will not impact the efficiency of prices.

For a more precise interpretation of prediction P2, the analysis will make a distinction between convertible bond arbitrage and other arbitrage activity (e.g., valuation shorts or exploitation of known autocorrelation). It may be reasonable to expect short selling due to general arbitrage activity to improve price efficiency; however, convertible bond arbitrageurs typically take their positions to hedge their bond positions and thus, stock price efficiency should not be affected.

In the empirical analysis, we use a variety of proxies for both liquidity and price efficiency. For liquidity, we examine: turnover, number of trades, the [Amihud \(2002\)](#) illiquidity measure, order imbalance (the absolute value of the difference between the number of buyer- and seller-initiated trades, classified based on [Lee and Ready, 1991](#)), quoted spread, quoted depth, and the ratio of spread to depth. High values for turnover, number of trades, and depth are interpreted as high liquidity. Low values of the [Amihud \(2002\)](#) measure, order imbalance, spread, and *spread/depth* are interpreted as high liquidity. The *spread/depth* measure is of interest because it reflects both the price (spread) and quantity (depth) aspects of stock quotes and can provide more insight than examining these measures separately. The ratio of spread to depth has also been used in prior work examining the impact of a derivatives market for the quality of the underlying stock market (see [Kumar, Sarin, and Shastri, 1998](#)). For stock price efficiency, we use: (1) the variance ratio, which compares stock price variances over different frequencies, where smaller deviations from one imply greater efficiency¹²; and (2) autocorrelation, where smaller magnitude of return autocorrelation is interpreted as greater

⁷ The risks and rewards of liquidity provision by convertible bond arbitrage hedge funds are studied in [Agarwal, Fung, Loon, and Naik \(2007\)](#).

⁸ This finding aids in defining windows used in our main analysis of the potential impact of convertible bond activity.

⁹ For example, see [Diamond and Verrecchia \(1987\)](#).

¹⁰ We thank the referee for suggesting this possibility.

¹¹ [Chakraborty and Yilmaz \(2006\)](#) suggest convertible bonds as a solution to adverse selection problems in the market for new securities when investors are uninformed.

¹² See [Lo and MacKinlay \(1988\)](#).

Table 1

Issuing firms and characteristics.

This table presents summary statistics for the sample of convertible bond issues between July 1993 and May 2006. *Market cap* is the issuing firm's equity market capitalization. *NYSE* and *Nasdaq* are dummy variables, indicating where the issuing firm is listed. *Debt/equity* is the ratio of long-term debt to equity market capitalization in the fiscal year prior to issuance. *Daily dollar volume* is the average daily dollar volume of the stock. *Beta* is the coefficient estimate of the regression of daily stock excess returns on CRSP value-weighted market excess return. *Issue size* is the face value of the convertible bond times its offer price. *Short interest* is the average monthly short interest. *Institutional holdings/shares outstanding* is the institutional holdings (by 13f institutions) divided by shares outstanding in the calendar year end prior to issuance. *Credit rating* is the bond rating issued by S&P.^a All daily and monthly measures are calculated using data from the six months ending one month prior to announcement of the issue.

Number of observations = 846.

	Mean	Median	Standard deviation
Market cap (\$ million)	4,687	1,179	13,457
NYSE	0.49	0	0.50
Nasdaq	0.51	1	0.50
Debt/equity	0.60	0.18	1.58
Daily dollar volume (\$ million)	42.58	12.67	103.10
Beta	1.34	1.27	0.75
Issue size (\$ million)	291.20	175.50	368.92
Issue size/market cap (%)	17.97	14.90	13.38
Short interest (000 shares)	5,497	2,152	14,753
Short interest/shares outstanding (%)	4.47	3.05	4.75
Institutional holdings/shares outstanding (%)	65.00	68.37	22.45
Credit rating	BB	BB-	

^aIf the bond is not rated by S&P, Moody's or Fitch rating is used, in that order, as available. Of 846 bonds 444 are rated by at least one of the three agencies. For calculating the mean and median credit rating, a number is assigned to each rating: best (AAA or Aaa) = 1, second best = 2, etc. Using this system, mean rating = 12.29, which lies between BB and BB- for S&P and Fitch (or Ba2 and Ba3 for Moody's), median = 13 (BB- or Ba3), standard deviation = 3.85. Only rated issues are included in the calculation.

efficiency.¹³ We also examine long-run stock returns following bond issue. The latter is a test of efficiency in that it asks whether the short-sales positions that we observe in the data would make money over various horizons.

4. Data and sample selection

4.1. Short interest and convertible debt issues

The initial sample consists of all convertible debt issues (public, private, and Rule 144a) by U.S. publicly traded firms for the period July 1993 through May 2006.¹⁴ Issue dates and other characteristics of the issues are from the SDC Global New Issues database and the Mergent Fixed Income Securities Database (FISD). We obtain monthly short-interest data directly from the NYSE and Nasdaq and match the short-interest data with the SDC data using ticker and date identifiers. Because the monthly short-interest files reflect short sales through three trading days (five for the first years of the sample) prior to the 15th of each month, we calculate a trade date

for each file and use that date to match to the SDC data.¹⁵ We then match these data to the Center for Research in Security Prices (CRSP)/Compustat tapes and NYSE TAQ Database. We also obtain data on institutional holdings from the Thomson Financial Institutional (13f) Holdings and analyst opinion from Institutional Brokers Estimate System (I/B/E/S). For inclusion in the final sample, we require non-missing data on short interest, all liquidity and efficiency measures, and all control variables such as institutional holdings, analyst opinion, and historical return volatility. This results in a final sample of 846 convertible bond issues.

Table 1 contains summary statistics. The issuing firms have a mean (median) market capitalization of \$4.7 (\$1.2) billion. The convertible bond issue sizes constitute significant proportion of equity value, with the mean (median) dollar value of proceeds equal to 18.0% (14.9%) of equity market capitalization. The firms for which we observe credit ratings are typically rated “junk,” with median Standard & Poor's (S&P) rating of BB-. In addition,

¹³ In unreported tests, we examined two additional efficiency measures: idiosyncratic volatility and *R*-squared. Results using these two measures are similar to the other efficiency measures. The distinction between idiosyncratic and systematic volatility is motivated by Bris et al. (2007). They interpret an observed low *R*-squared as evidence of efficiency. Similarly, we interpret an increase in idiosyncratic volatility as evidence of improved price efficiency because it suggests that more firm-specific information is incorporated into prices.

¹⁴ We begin the analysis in 1993 because NYSE Trade and Quote (TAQ) data are used to construct some of the liquidity and price-efficiency measures.

¹⁵ It is critical to correctly match the short-interest dates to the issue dates. The monthly short-interest data are based on short interest as of trade dates that occur during the middle of the month at non-constant days across months (due to settlement). Following the documentation from the short-interest files that we received from Nasdaq and the NYSE, we define the cutoff trade date for a given month as: five trading days before the 15th (or the preceding trading day if the 15th is not a trading day), through June 1995; and three trading days before the 15th after June 1995. If a bond is issued before the cutoff trade date of a given month, the short-interest data file for that month is matched to the issue month. Otherwise, the short-interest data for the following month is matched to the issue month. This algorithm is consistent with Bechmann (2004).

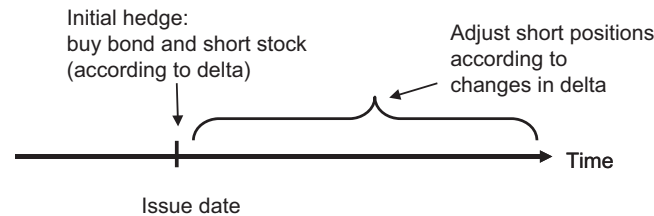


Fig. 1. Convertible bond arbitrage delta-neutral hedging. This figure presents the timing of a delta-neutral hedge, in which the convertible bond arbitrageur buys a convertible bond and shorts the underlying stock according to the delta of the option embedded in the bond. As stock price changes over time, the position must also be adjusted in order to maintain a delta-neutral hedge. The arbitrageur will sell short additional stock when the stock price increases and buy/cover the short position as stock price declines.

the sample consists of about the same number of NYSE and Nasdaq issuers. We do observe some short selling in these stocks prior to issuance and announcement, with mean (median) short interest during the six months prior to announcement equal to 4.5% (3.1%) of shares outstanding. This is relevant, as arbitrageurs are likely to be attracted to stocks for which a relatively liquid short selling market exists.

4.2. Proxy for the presence of the convertible bond arbitrage strategy

Our proxy for the presence of the convertible bond arbitrageurs is the change in short-interest intensity (“ ΔSI ”) during the month of the convertible bond issue. As discussed in Section 1, the typical convertible bond arbitrage strategy employs delta-neutral hedging, and consists of two parts. The arbitrageur initially buys the convertible bond and sells short the underlying equity at the current delta. Next, if the price of the stock increases, the arbitrageur adds to the short position because the delta has increased. Similarly, when the stock price declines, the arbitrageur buys stock due to the decrease in delta. Fig. 1 provides a timeline. Importantly, dynamic hedging is expected to improve liquidity because aggregate equity market trading demand is expected to move in the opposite direction (see Chordia, Roll, and Subrahmanyam, 2002). We do not directly observe arbitrageurs’ dynamic shorting and covering transactions. Fortunately, their initial arbitrage positions can be reasonably captured since it is straightforward to identify the dates on which they are established. We expect the shares initially shorted by convertible bond arbitrageurs to be a good proxy for post-issue dynamic hedging activity because it measures the presence of arbitrageurs in the equity market. We initially define two measures to proxy for arbitrage activity:

- ΔSI_Shrout_t is the change in short interest (number of shares) during period t , scaled by total shares outstanding in period $t - 1$. The change in short interest is the difference between short interest in month t and short interest during month $t - 1$.
- ΔSI_Issue_t is the dollar value change in short interest during the period t , divided by issue proceeds.

The first measure, ΔSI_Shrout_t provides a measure of the relative importance of the new arbitrageurs in the

market for the stock. The second measure, ΔSI_Issue_t , is related to issue characteristics—namely, the amount of short selling activity as a fraction of the issue size (which can be directly linked to hedging activity).

We do not expect convertible bond arbitrageurs to short before the bond issue due to the risk associated with having an unhedged stock position. Similarly, we do not expect them to delay the initial hedge since the bond position will also give them unhedged exposure to equity risk. Fig. 2 reports means and medians of the ΔSI measures during months -6 to $+6$ relative to the issue date. Consistent with our ex ante expectation, the figures show that we are capturing an increase in short interest related to the issue. The median increase in short interest relative to shares outstanding at issue month 0 (ΔSI_Shrout_t) is 1.7%. The median dollar value increase in short interest relative to issue size at issue month 0 (ΔSI_Issue_t), is 13.1%.

As shown in the Fig. 2, both ΔSI_Shrout_t and ΔSI_Issue_t capture similar variation in short selling activity. The main analysis uses ΔSI_Shrout_t as a proxy for convertible bond arbitrage activity due to our interest in the implications of convertible bond arbitrage for the market for the underlying stock.¹⁶ In addition, we focus on the issue month 0, given the large increase in short interest that occurs at that time. For notational convenience, we denote the proxy for convertible bond arbitrage activity as ΔSI (i.e., we drop both $_Shrout$ and the t subscript).¹⁷

Fig. 3 provides a description of the time series of convertible bond issuance and the size of convertible bond arbitrage hedge funds. Issuance has steadily increased over time. We have also seen growth in the total assets managed by convertible bond arbitrage hedge funds, which is consistent with a role for hedge funds as suppliers of capital as discussed in Section 1. We also examined the time series of changes in short interest during the sample period and, not surprisingly, observe

¹⁶ However (in unreported tests) we have replicated the analysis using ΔSI_Issue_t . All liquidity results are qualitatively similar (but weaker). The efficiency results are almost identical.

¹⁷ Though it is true that short sellers can also short due to private information (see, e.g., Christophe et al., 2004, for short selling prior to earnings announcements) or other types of arbitrage activity, the fact that we capture the increase in shorting over a relatively short horizon relative to the bond issue date suggests that our ΔSI measures are, in large part, capturing convertible bond arbitrage. We explicitly distinguish valuation shorting from arbitrage shorting in an analysis of short selling near announcement of the issue versus the actual issue date (see the discussion in Sections 5.4).



Fig. 2. Mean and median change in short interest (number of observations = 846). The charts show the mean and median change in short interest during the event window (months –6 to +6). $\Delta SI_ \% Issue_t$ is the dollar value of the change in short interest during month t , divided by issue size. That is: difference between short interest in month t and short interest in month $t - 1$, times the closing stock price in month t , divided by issue size (face value of the convertible bond times offer price). $\Delta SI_ \% Shrout_t$ is the change in short interest during month t divided by the number of shares outstanding in month $t - 1$. The sample period is from July 1993 to May 2006.

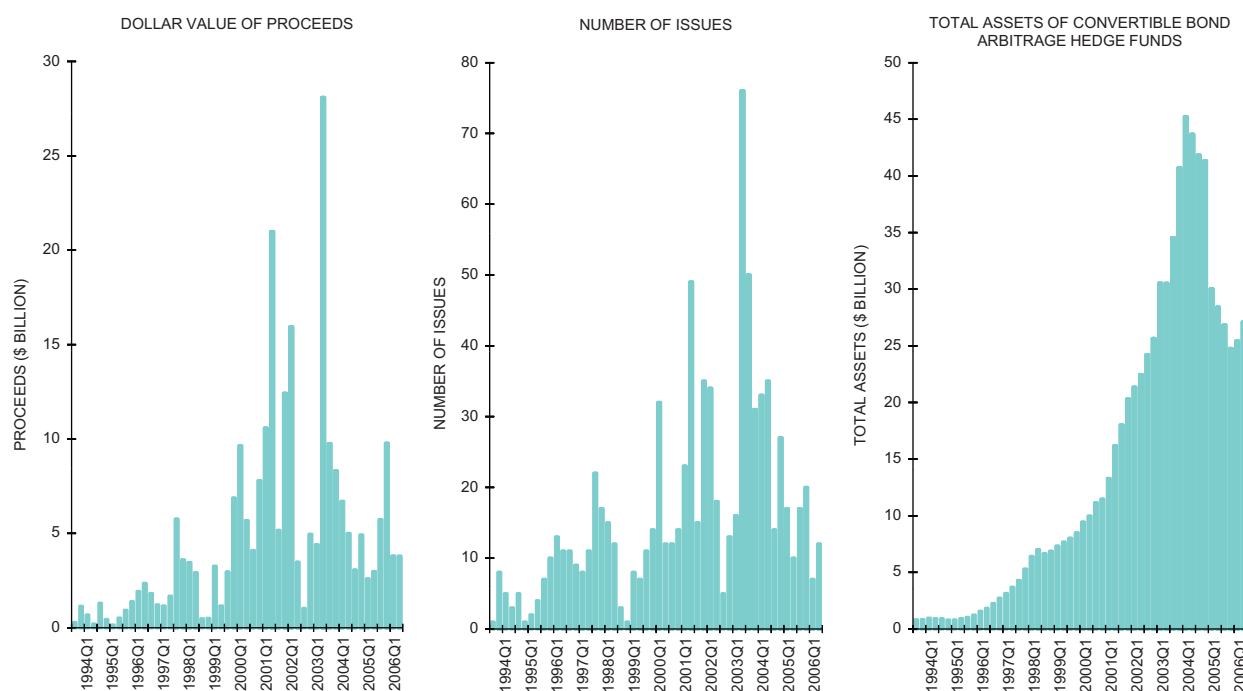


Fig. 3. Dollar value of proceeds, number of issues, and hedge fund size. The charts show the total dollar value of proceeds from convertible bonds, the number of convertible bond issues, and the total assets of convertible bond arbitrage hedge funds from 1993 Q3 to 2006 Q2.

significant time-series variation in the data. Given this observation and findings in the literature of distinct time-series patterns in short interest (see, e.g., Lamont and Stein, 2004), we include year and month fixed effects in all cross-sectional regression specifications.

In the main analysis of changes in liquidity and stock price efficiency we examine a relatively short time

horizon in order to isolate the impact of the convertible bond arbitrage strategy.¹⁸ Specifically, we examine

¹⁸ Convertible bonds often have call provisions; however, beginning with Ingersoll (1977) the empirical evidence has suggested that firms call too late. Further, callability should minimally impact our study over

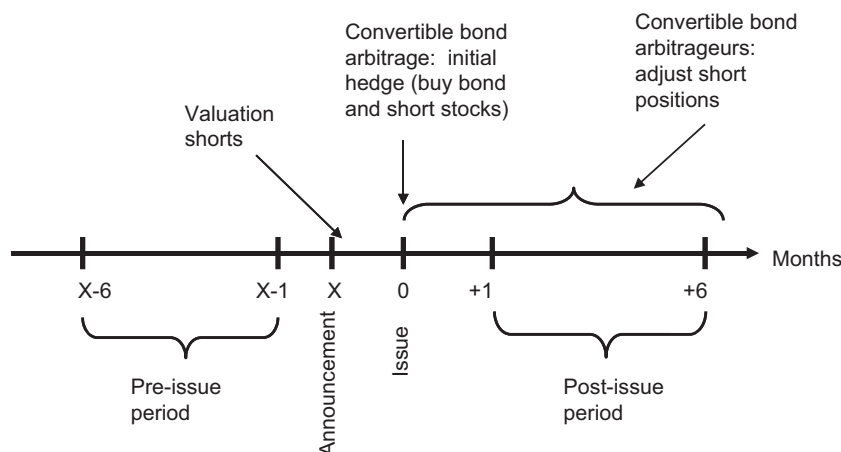


Fig. 4. Time intervals for short interest, liquidity, and stock price variables. This figure illustrates the time horizons over which changes in short interest, liquidity, and stock price-efficiency variables are measured. The “post-issue period” is the six-month (120 trading days) period beginning one month (20 trading days) after the bond issue (Day 0). The “pre-issue period” is the six-month (120 trading days) period ending one month (20 trading days) prior to the announcement of the convertible bond issue (Day X).

changes in liquidity and efficiency from the pre-issue period to the post-issue period. The “post-issue” period is defined as the six months (120 trading days) beginning one month (20 trading days) following the bond issue. The “pre-issue” period is defined as the six months (120 trading days) ending one month (20 trading days) prior to the announcement of the bond. We skip the two months immediately surrounding announcement and issuance in calculating pre- and post-issue liquidity and stock price measures.¹⁹ We do this to avoid the direct impact of traders establishing valuation and/or initial arbitrage positions. A timeline defining these windows of interest is given in Fig. 4.

As Fig. 4 shows, the initial delta hedge associated with buying the bond and shorting the stock is implemented at issuance. After the issue, arbitrage activity associated with dynamic delta-hedging occurs. Note that valuation shorting is expected to occur near the announcement (not the issuance) of the bond. Note also that the liquidity changes are expected to occur over an extended time period via the dynamic arbitrage activity following issuance. Because this hedging activity is unobservable, we capture the initial positions taken by the arbitrageurs (at $t = 0$) to proxy for their presence in the post-issue market for the stock. We relate this measure to post-issue liquidity changes.

5. Convertible bond arbitrage, liquidity, and stock prices

In this section, we examine links between changes in short interest near issuance and equity market characteristics.

5.1. Summary of firm characteristics, by ΔSI portfolio

Table 2 provides summary statistics of all of the sample firms prior to issuance.²⁰ The column “All” describes the full sample of issuers. We also divide the sample into four portfolios based on the change in short interest at issue (the ΔSI measure) in order to provide some insight into the types of issuers for which the convertible bond arbitrage strategy is most evident. Portfolio 1(4) corresponds to the smallest (largest) short-interest change. There are several relevant observations from Panel A of the table. First, Nasdaq stocks see the largest ΔSI following issuance. Second, small issuers and private issues experience higher ΔSI in their underlying stocks. Third, convertible bond arbitrage activity is higher in stocks that have a high pre-issue short interest, indicating that arbitrageurs choose issues where they believe they will have the ability to short the stock. Finally, as would be expected if convertible bond arbitrageurs are shorting shares to manage equity risk, the amount of short selling at issuance is positively and significantly related to the conversion ratio (number of shares into which the bond can be converted).

Panel B of Table 2 reports stock market liquidity measures. The number of trades, dollar volume, the Amihud (2002) illiquidity measure, order imbalance, spread, and spread-to-depth measures all indicate that

(footnote continued)

the six-month horizon because callable bonds often have call protection periods, generally greater than six months. See, e.g., Asquith (1995).

¹⁹ The bond announcement dates are hand-collected. Of 846 issues, the newswires suggest that 28 are issued before announcement (23 of these are private issues). For these observations, the pre- and post-periods are defined as: 120 trading days ending one month prior to issuance and beginning one month following announcement, respectively. Our results are not affected if we instead drop these 28 issues from the sample.

²⁰ All measures are calculated using daily or monthly data from the six months (two months for Intraday AR(1)) ending one month prior to announcement of the bond issue.

Table 2Issuing firm characteristics, ΔSI portfolio sorts.

This table presents firm characteristics prior to issuance. Portfolios are based on issue period change in short interest (ΔSI), the proxy for convertible bond arbitrage activity. ΔSI is the change in short interest (from the month prior to issuance) divided by the number of shares outstanding in the month prior to issuance.

In Panel A, *NYSE* and *Nasdaq* are dummy variables, indicating where the issuing firm is listed. *Public* is one if the convertible bond is a public offering, zero if private. *Market cap* is the equity market capitalization. *Short interest/shares outstanding* is the average monthly short interest divided by shares outstanding prior to issuance. *Institutional holdings/shares outstanding* is the institutional holdings (by 13f institutions) divided by shares outstanding at calendar year end prior to issuance. *Conversion ratio* is the number of shares of common stock that could be obtained by converting one bond. *Conversion premium* is the amount (in percent) by which the conversion price exceeds the market value of the common stock at issuance.

In Panel B, *Turnover* is the average daily volume divided by shares outstanding. *Number of trades* is the average daily number of stock transactions on the firm's primary exchange. *Dollar volume* is the average daily dollar stock volume. *Amihud* is the average ratio of daily absolute return to dollar volume (Amihud, 2002). *OIBNUM* is the average daily absolute difference between the numbers of buyer- and seller-initiated trades divided by their sum. *Dollar spread* and *Percentage spread* are the difference between bid and ask quotes (time-weighted), expressed as dollars and percentage of bid-ask midpoint, respectively. *Total depth/shares outstanding* is the sum of bid and ask quoted depths divided by shares outstanding. *Dollar spread/dollar depth* is the ratio of quoted spread to quoted depth, both expressed in dollars.

In Panel C, *Return* and *Standard deviation of return* are the mean and standard deviation of daily stock return, respectively. *Idiosyncratic volatility* and *R-squared* are, respectively, the standard deviation of residuals and *R*-squared from the regression of daily stock excess return on CRSP value-weighted market excess return. *Beta* is the coefficient estimate of the same regression. *Daily AR(1)* and *Intraday AR(1)* are the first-order autocorrelation of returns, calculated using daily returns and 30-minute interval returns, respectively. *Variance ratio* (5) is the five-day variance ratio in Lo and MacKinlay (1988).

All measures in Panels B and C are calculated using daily or monthly data from the six months (two months for *Intraday AR(1)*) ending one month prior to announcement. The last two columns show the mean measures of Portfolio 4 minus Portfolio 1 and the corresponding industry- and time-clustered *t*-statistics. *, **, and *** denote 10%, 5%, and 1% significance, respectively. The sample period is from July 1993 to May 2006.

Number of observations = 846.

	Portfolios based on ΔSI						
	All	P1	P2	P3	P4	P4 – P1	t-stat
	(Smallest)			(Largest)			
<i>Panel A: Firm and convertible bond characteristics</i>							
NYSE	0.505	0.626	0.552	0.474	0.349	–0.277***	(–5.59)
Nasdaq	0.495	0.360	0.448	0.526	0.646	0.286***	(5.79)
Public	0.252	0.280	0.278	0.265	0.184	–0.096**	(–2.29)
log Market cap	21.076	21.720	21.402	20.779	20.404	–1.316***	(–9.54)
Short interest/shares outstanding (%)	4.466	4.476	3.851	4.311	5.185	0.709***	(13.95)
Institutional holdings/shares outstanding (%)	65.002	65.171	65.768	63.860	65.205	0.034	(1.52)
Conversion ratio	43.918	37.216	33.681	50.348	54.426	17.211***	(2.80)
Conversion premium (%)	35.062	34.624	36.190	37.043	32.400	–2.223	(–0.61)
<i>Panel B: Liquidity measures</i>							
log Turnover	–4.949	–5.103	–4.955	–4.941	–4.799	0.304***	(3.47)
log Number of trades	5.987	6.220	6.135	5.849	5.744	–0.476***	(–3.28)
log Dollar volume	16.125	16.615	16.446	15.836	15.603	–1.013***	(–5.95)
log Amihud	–20.585	–21.240	–20.903	–20.267	–19.929	1.311***	(9.27)
OIBNUM (%)	16.789	15.909	16.302	17.201	17.741	1.832**	(2.37)
Dollar spread	0.125	0.114	0.122	0.124	0.139	0.026*	(1.88)
Percentage spread (%)	0.538	0.440	0.433	0.618	0.662	0.222***	(3.44)
Total depth/shares outstanding (%) $\times$ 1000	8.366	7.185	6.859	8.904	10.512	3.327**	(2.19)
Dollar spread/dollar depth (%) $\times$ 1000	50.395	38.832	40.918	53.708	68.605	29.773***	(4.70)
<i>Panel C: Return and price-efficiency measures</i>							
Return (%)	0.238	0.166	0.269	0.256	0.261	0.094***	(2.61)
Standard deviation of return (%)	3.572	3.053	3.333	3.726	4.175	1.122***	(6.27)
Idiosyncratic volatility (%)	3.210	2.699	2.997	3.344	3.799	1.100***	(6.81)
R-squared (%)	18.482	21.655	17.603	18.535	16.149	–5.505***	(–3.84)
Beta	1.334	1.257	1.317	1.366	1.396	0.139*	(1.88)
Daily AR(1) (%)	–0.965	–2.324	–1.145	0.174	–0.567	1.757	(1.56)
Intraday AR(1) (%)	0.048	–0.528	–0.153	0.336	0.536	1.063	(1.26)
Variance ratio (5)	1.035	1.022	1.014	1.060	1.045	0.023	(0.99)

stocks in the smallest ΔSI portfolio are more liquid. However, as noted above, firms in the high ΔSI portfolio tend to be smaller, making the direct comparison of the level of liquidity measures inappropriate. We therefore focus on *changes* in liquidity in the main analysis, and control for pre-issue liquidity level and change in firm size in the regressions.

Panel C of Table 2 presents descriptive statistics on a variety of return and price-efficiency measures. We observe higher convertible bond arbitrage activity in stocks with higher average returns and standard deviation of returns, as well as higher betas, higher idiosyncratic volatility, and lower *R*-squared parameters (estimated from a market model regression). We also calculate

autocorrelation of returns and variance ratios (see [Lo and MacKinlay, 1988](#)), which we use as measures of the degree of price efficiency. Daily and intraday AR(1) parameters are calculated using daily returns and 30-minute interval returns, respectively. From the table we do not observe a significant relationship between changes in short interest and these efficiency measures. This suggests that stock price efficiency is not an important factor in convertible bond arbitrage (as would be expected, if equity positions are taken primarily to hedge equity risk).

5.2. Impact of convertible bond arbitrage on liquidity and prices

5.2.1. Average changes, by ΔSI portfolio

In [Table 3a](#), we present the changes in firm characteristics, sorted by portfolios based on change in short interest at issuance (ΔSI). The column “All” includes data for the full sample of issuers and is presented for comparability to previous studies of changes in liquidity and stock return volatility following the introduction of derivatives markets. As in [Table 2](#), the portfolio sort groups issuers into Portfolios 1 through 4, which correspond to the smallest through largest issue-month short-interest changes. All changes in liquidity and stock price variables are defined as the post-issue period mean minus the pre-issue period mean. Along with changes in short interest, we measure changes in the following liquidity proxies: share turnover, number of trades, dollar volume, the illiquidity measure developed by [Amihud \(2002\)](#), order imbalance (absolute difference between buyer- and seller-initiated trades), and time-weighted average quotes.²¹ The [Amihud \(2002\)](#) measure is a proxy for [Kyle's \(1985\) \$\lambda\$](#) and is defined as absolute return divided by dollar volume.

We find strong evidence of an increase in liquidity based on all measures following issuance, with the exception of quoted depth (which indicates a decrease in liquidity).²² Consistent with the prediction (P1), these improvements increase systematically with the proxy for arbitrage activity, ΔSI . For example, the change in (log) turnover for the largest ΔSI portfolio is 0.31 higher than that for the smallest ΔSI portfolio. Importantly, because we link liquidity changes to ΔSI , we provide direct evidence of the impact of arbitrageurs on liquidity. Prior literature on the impact of derivatives markets on stock markets shows only average changes in these variables (see, e.g., [Mayhew, 2000](#), for a survey) and does not examine systematic relationships with arbitrage activity.

For stock prices and efficiency, we examine the following measures: average daily returns, standard deviation of daily returns, idiosyncratic volatility, R -squared, beta, AR(1) parameters, and variance ratios. In regression analysis, we rely on the latter two variables to capture changes in efficiency. If arbitrageurs impact stock price efficiency, then we would expect decreases in

return predictability, as captured by the AR(1) parameters. Further, the variance ratio ([Lo and MacKinlay, 1988](#)) captures the extent to which stock prices follow a random walk. The standard deviation of returns is included in [Table 3a](#) so that we can compare the results with the empirical regularity of decreases in volatility following the introduction of options markets. Beta and R -squared are motivated by [Bris et al. \(2007\)](#).

The portfolio sorts presented in Panel B of [Table 3a](#) suggest that the impact of convertible bond arbitrage on stock price efficiency is very weak. Consistent with prior work, we do find an average decrease in both total return volatility and the idiosyncratic component of volatility following convertible bond issuance. However, we do not find evidence that these average declines vary systematically with short selling activity. That is, there is no evidence that arbitrage is what is driving the declines. Average returns decrease after issuance and these decreases are higher for the highest ΔSI portfolios, consistent with the observation that returns decrease following announcement of convertible bond issues. Beta and R -squared both increase. We do not observe significant changes in the AR(1) parameters or variance ratios. Across ΔSI portfolios, the only systematic variation that we observe is in returns and beta. Taken together, the results in Panel B of [Table 3a](#) do not indicate an impact of convertible bond arbitrage on stock price efficiency. Regression analysis (below) will further investigate these findings.

5.2.2. Control sample

It is possible that the liquidity results in [Table 3a](#) are being driven by market-wide changes in liquidity, rather than convertible bond arbitrage activity. To examine this potentially important issue, we attempt to isolate changes due to the convertible bond event by analyzing the measures in [Table 3a](#) for a set of control firms. In [Table 3b](#), we examine the possibility that the results are driven by the market-wide trends in general, rather than convertible bond arbitrage. To do this, we match firms in the sample based on industry, exchange, size, market-to-book, and turnover before issuance. To identify the control firms, we begin with all firms in the CRSP/Compustat database. From this initial sample the sequence of identifying control firms proceeds as follows:

- First, firms that have issued any convertible debt during years -1 to $+1$ relative to issue are eliminated from the universe of potential control firms.
- Second, control firms must trade on the same exchange as issuing firms (e.g., NYSE issuers are matched only to NYSE firms). This is done to eliminate potential problems associated with exchange-related trends in liquidity and stock price movements.
- Third, control firms must be in the issuing firm's industry, based on [Fama and French \(1997\)](#) industry code.²³

²¹ We also examine opening quotes. Results are qualitatively similar.

²² However, regression analysis ([Table 4](#)) shows that quoted depth increases with arbitrage activity, after we control for other variables.

²³ After filtering, the mean (median) number of valid control firms for each issuing firm is 166 (101) and there are at least two potential control firms for each issuer.

Table 3

(a) Changes in firm characteristics.

This table presents the changes (from pre- to post-issue) in firm characteristics, by portfolios based on issue period change in short interest (ΔSI), the proxy for convertible bond arbitrage activity. ΔSI is the change in short interest (from the month prior to issuance) divided by the number of shares outstanding in the month prior to issuance. Changes in firm characteristics (Δ) are defined as the average measure in post-issue period minus the pre-issue period average measure. The “pre-issue period” is defined as the six months (two months for *Intraday AR*(1)) ending one month prior to announcement. The “post-issue period” is the six months (two months for *Intraday AR*(1)) starting one month after issuance.

In Panel A, *Short interest/shares outstanding* is the average monthly short-interest level divided by shares outstanding. *Turnover* is the average daily volume divided by shares outstanding. *Number of trades* is the average daily number of stock transactions on the firm's primary exchange. *Dollar volume* is the average daily dollar stock volume. *Amihud* is the average ratio of daily absolute return to dollar volume (Amihud, 2002). *OIBNUM* is the average daily absolute difference between the numbers of buyer- and seller-initiated trades divided by their sum. *Dollar spread* and *Percentage spread* are the difference between bid and ask quotes (time-weighted), expressed as dollars and percentage of bid–ask midpoint, respectively. *Total depth/shares outstanding* is the sum of bid and ask quoted depths divided by shares outstanding. *Dollar spread/dollar depth* is the ratio of quoted spread to quoted depth, both expressed in dollars.

In Panel B, *Return* and *Standard deviation of return* are the mean and standard deviation of daily stock return, respectively. *Idiosyncratic volatility* and *R-squared* are, respectively, the standard deviation of residuals and *R-squared* from the regression of daily stock excess return on CRSP value-weighted market excess return. *Beta* is the coefficient estimate of the same regression. $|Daily AR(1)|$ and $|Intraday AR(1)|$ are the absolute value of first-order autocorrelation of returns, calculated using daily returns and 30-minute interval returns, respectively. $|Variance ratio(5) - 1|$ is the absolute deviation of the five-day variance ratio in Lo and MacKinlay (1988) from one.

The last column shows the mean measures of Portfolio 4 minus Portfolio 1. Industry- and time-clustered *t*-statistics of the changes and differences are in parentheses. *, **, and *** denote 10%, 5%, and 1% significance, respectively. The sample period is from July 1993 to May 2006.

Number of observations = 846.

	Portfolios based on ΔSI					
	All	P1 (Smallest)	P2	P3	P4 (Largest)	P4 – P1
<i>Panel A: Changes in liquidity measures</i>						
Δ Short interest/shares outstanding (%)	2.371*** (17.90)	0.181 (1.02)	1.233*** (8.44)	2.598*** (11.55)	5.465*** (19.27)	5.284*** (15.81)
Δ log Turnover	0.239*** (15.06)	0.079*** (3.25)	0.217*** (7.27)	0.272*** (9.41)	0.387*** (11.29)	0.308*** (7.35)
Δ log Number of trades	0.325*** (15.74)	0.195*** (6.52)	0.343*** (9.07)	0.336*** (9.67)	0.426*** (10.57)	0.231*** (4.60)
Δ log Dollar volume	0.441*** (15.88)	0.241*** (6.75)	0.436*** (9.37)	0.510*** (10.77)	0.574*** (9.13)	0.333*** (4.60)
Δ log Amihud	−0.458*** (−17.02)	−0.330*** (−8.55)	−0.475*** (−11.52)	−0.510*** (−11.20)	−0.519*** (−8.05)	−0.188*** (−2.51)
Δ OIBNUM (%)	−1.041*** (−7.07)	−0.893*** (−3.72)	−0.924*** (−3.26)	−0.961*** (−3.57)	−1.385*** (−4.58)	−0.492 (−1.27)
Δ Dollar spread	−0.021*** (−7.40)	−0.022*** (−4.30)	−0.017*** (−3.21)	−0.019*** (−4.42)	−0.027*** (−4.70)	−0.006 (−0.77)
Δ Percentage spread (%)	−0.134*** (−11.48)	−0.081*** (−5.36)	−0.105*** (−9.36)	−0.188*** (−5.52)	−0.160*** (−6.57)	−0.079*** (−2.77)
Δ Total depth/shares outstanding (%) $\times 1000$	−0.712** (−1.97)	−0.287 (−0.95)	−0.192 (−0.35)	−1.903 (−1.58)	−0.467 (−0.96)	−0.180 (−0.31)
Δ Dollar spread/dollar depth (%) $\times 1000$	−13.970*** (−14.06)	−9.260*** (−5.21)	−11.230*** (−6.18)	−14.900*** (−7.46)	−20.470*** (−9.61)	−11.210*** (−4.04)
<i>Panel B: Changes in return and price-efficiency measures</i>						
Δ Return (%)	−0.201*** (−11.70)	−0.128*** (−4.26)	−0.230*** (−7.76)	−0.214*** (−6.83)	−0.232*** (−5.91)	−0.105** (−2.12)
Δ Standard deviation of return (%)	−0.250*** (−4.20)	−0.341*** (−3.98)	−0.183* (−1.94)	−0.356*** (−3.49)	−0.123 (−0.96)	0.219 (1.42)
Δ Idiosyncratic volatility (%)	−0.275*** (−5.37)	−0.339*** (−4.26)	−0.209** (−2.55)	−0.358*** (−3.88)	−0.193* (−1.66)	0.146 (1.04)
Δ R-squared (%)	2.309*** (4.48)	1.678* (1.88)	1.790** (2.10)	2.152** (2.24)	3.612*** (3.66)	1.934 (1.45)
Δ Beta	0.083*** (3.06)	0.046 (1.21)	0.025 (0.47)	0.080* (1.67)	0.180*** (3.41)	0.135** (2.07)
$\Delta Daily AR(1) $ (%)	−0.295 (−0.87)	0.077 (0.11)	−0.419 (−0.58)	−0.431 (−0.66)	−0.406 (−0.61)	−0.483 (−0.50)
$\Delta Intraday AR(1) $ (%)	2.521 (1.34)	9.973 (1.34)	0.856 (1.23)	−0.276 (−0.67)	−0.447 (−0.77)	−10.420 (−1.40)
$\Delta Variance ratio(5) - 1 $ (%)	−1.102 (−1.43)	−0.584 (−0.41)	−0.911 (−0.60)	−2.490 (−1.47)	−0.428 (−0.28)	0.156 (0.07)

Table 3 (continued)

(b) Control firm results: changes in firm characteristics.
This table shows the changes (from pre- to post-issue) in issuing firm characteristics *minus* the changes in control firm characteristics. Portfolios are based on issue period change in short interest (ΔSI), the proxy for convertible bond arbitrage activity. ΔSI is the change in short interest (from the month prior to issuance) divided by the number of shares outstanding in the month prior to issuance. Control firms are matched based on size, book-to-market, and turnover before issuance, exchange, and industry. Changes in firm characteristics (Δ) are defined as the average measure in post-issue period minus the pre-issue period average measure. The “pre-issue period” is defined as the six months (two months for *Intraday AR*(1)) ending one month prior to announcement. The “post-issue period” is the six months (two months for *Intraday AR*(1)) starting one month after issuance.

In Panel A, *Short interest/shares outstanding* is the average monthly short interest of the stock divided by shares outstanding. *Turnover* is the average daily volume divided by shares outstanding. *Number of trades* is the average daily number of stock transactions on the firm's primary exchange. *Dollar volume* is the average daily dollar stock volume. *Amihud* is the average ratio of daily absolute return to dollar volume (Amihud, 2002). *OIBNUM* is the average daily absolute difference between the numbers of buyer- and seller-initiated trades divided by their sum. *Dollar spread* and *Percentage spread* are the difference between bid and ask quotes (time-weighted), expressed as dollars and percentage of bid–ask midpoint, respectively. *Total depth/shares outstanding* is the sum of bid and ask quoted depths divided by shares outstanding. *Dollar spread/dollar depth* is the ratio of quoted spread to quoted depth, both expressed in dollars.

In Panel B, *Return* and *Standard deviation of return* are the mean and standard deviation of daily stock return, respectively. *Idiosyncratic volatility* and *R-squared* are, respectively, the standard deviation of residuals and *R-squared* from the regression of daily stock excess return on CRSP value-weighted market excess return. *Beta* is the coefficient estimate of the same regression. $|Daily\ AR(1)|$ and $|Intraday\ AR(1)|$ are the absolute value of first-order autocorrelation of returns, calculated using daily returns and 30-minute interval returns, respectively. $|Variance\ ratio(5) - 1|$ is the absolute deviation of the five-day variance ratio in Lo and MacKinlay (1988) from one.

The last column shows the mean measures of Portfolio 4 minus Portfolio 1. Industry- and time-clustered *t*-statistics of the differences are in parentheses.

*, **, and *** denote 10%, 5%, and 1% significance, respectively. The sample period is from July 1993 to May 2006.

Number of observations = 846.

	Portfolios based on ΔSI					
	All	P1 (Smallest)	P2	P3	P4 (Largest)	P4 – P1
<i>Panel A: Liquidity measures (changes in issuing firm minus changes in control firm)</i>						
Δ Short interest/shares outstanding (%)	2.462*** (12.29)	0.252 (1.22)	0.875*** (3.17)	3.003*** (5.80)	5.726*** (16.21)	5.473*** (13.38)
Δ log Turnover	0.209*** (10.09)	0.048 (1.40)	0.184*** (4.44)	0.223*** (5.39)	0.379*** (8.84)	0.331*** (6.04)
Δ log Number of trades	0.162*** (7.66)	0.041 (1.13)	0.185*** (4.36)	0.134*** (3.13)	0.289*** (6.41)	0.248*** (4.30)
Δ log Dollar volume	0.321*** (10.34)	0.076 (1.53)	0.319*** (5.61)	0.356*** (5.46)	0.531*** (7.57)	0.455*** (5.29)
Δ log Amihud	–0.334*** (–11.45)	–0.123** (–2.46)	–0.317*** (–6.13)	–0.351*** (–5.74)	–0.545*** (–8.12)	–0.422*** (–5.04)
Δ OIBNUM (%)	–0.370* (–1.71)	0.233 (0.58)	–0.643 (–1.48)	–0.370 (–0.85)	–0.696* (–1.65)	–0.929 (–1.60)
Δ Dollar spread	0.001 (0.26)	0.001 (0.18)	0.001 (0.17)	0.005 (0.72)	–0.003 (–0.47)	–0.004 (–0.47)
Δ Percentage spread (%)	–0.045** (–2.42)	0.031 (1.32)	–0.063* (–1.81)	–0.069 (–1.62)	–0.077* (–1.82)	–0.107** (–2.23)
Δ Total depth/shares outstanding (%) $\times 1000$	0.569 (0.48)	1.482 (0.78)	–0.942 (–0.26)	–0.145 (–0.09)	1.880 (1.05)	0.398 (0.15)
Δ Dollar spread/dollar depth (%) $\times 1000$	–2.539 (–1.30)	6.465* (1.69)	–1.668 (–0.62)	0.931 (0.19)	–15.826*** (–3.96)	–22.291*** (–4.04)
<i>Panel B: Return and price-efficiency measures (changes in issuing firm minus changes in control firm)</i>						
Δ Return (%)	–0.102*** (–4.65)	–0.032 (–0.93)	–0.067 (–1.30)	–0.152*** (–4.17)	–0.155*** (–3.41)	–0.124** (–2.17)
Δ Standard deviation of return (%)	–0.130* (–1.77)	–0.079 (–0.90)	0.032 (0.19)	–0.358*** (–3.48)	–0.118 (–0.64)	–0.039 (–0.19)
Δ Idiosyncratic volatility (%)	–0.142* (–1.94)	–0.085 (–0.94)	0.012 (0.07)	–0.354*** (–3.23)	–0.142 (–0.76)	–0.057 (–0.28)
Δ R-squared (%)	0.757 (1.42)	0.165 (0.16)	0.588 (0.61)	1.014 (1.00)	1.259 (1.09)	1.094 (0.71)
Δ Beta	0.002 (0.07)	–0.043 (–1.03)	0.006 (0.09)	–0.058 (–1.10)	0.104 (1.60)	0.147* (1.90)
$\Delta Daily\ AR(1) $ (%)	–0.020 (–0.04)	1.108 (1.16)	–0.671 (–0.66)	–0.418 (–0.46)	–0.097 (–0.09)	–1.206 (–0.86)
$\Delta Intraday\ AR(1) $ (%)	4.152 (1.40)	2.691 (0.32)	6.651 (1.22)	8.521 (1.35)	–1.240 (–1.52)	–3.930 (–0.46)
$\Delta Variance\ ratio(5) - 1 $ (%)	0.026 (0.02)	1.652 (0.89)	–0.302 (–0.15)	–1.506 (–0.63)	0.259 (0.11)	–1.392 (–0.45)

- Finally, for the remaining sample of potential control firms, we assign a score based on turnover, market capitalization, and book-to-market during year $t - 1$, where t is the issue year. For each potential control firm we calculate a

$$\text{score} = \left[\text{abs} \left(\frac{\text{turnover}_{t-1}}{\text{issuer turnover}_{t-1}} - 1 \right) + \text{abs} \left(\frac{\text{market cap}_{t-1}}{\text{issuer market cap}_{t-1}} - 1 \right) + \text{abs} \left(\frac{\text{book-to-market}_{t-1}}{\text{issuer book-to-market}_{t-1}} - 1 \right) \right].$$

The firm with the lowest score (i.e., average distance along these three dimensions from the issuing firm) is chosen as the control firm.²⁴

Results from the control sample are reported in Table 3b. All results in Table 3b are presented as *differences* between the issuers and control firms. From the table, it is clear that the liquidity results are robust to measuring the changes relative to control firms. For stock price-efficiency measures we do not observe significant differences between the issuing firms and control firms, suggesting little or no role in stock price efficiency for convertible bond arbitrageurs. If convertible bond arbitrageurs take positions mainly to exploit mispricing in the bond (and not the stock), then this would be expected. Because it is important to control for market-wide effects, the change variables in the analysis henceforth are presented as deviations from control firms.

5.2.3. Regression analysis

We use an event study methodology to further characterize the relationships among convertible bond arbitrage, liquidity, and stock prices. These tests are more restrictive than the tests based on portfolio sorts in Tables 3a and b; however, we would like to explicitly control for factors other than ΔSI , issue period change in short-interest intensity. We use regression analysis to estimate the impact of convertible bond arbitrage and other stock characteristics of issuing firm i on changes in liquidity and price-efficiency measures.

$\Delta \text{Liquidity}_i$ or $\Delta \text{Efficiency}_i$

$$\begin{aligned} &= \alpha + \beta_1 \Delta SI_i + \beta_2 \Delta \text{Market Cap}_i + \beta_3 \Delta \text{Volatility}_i \\ &+ \beta_4 \Delta \text{Institutional Holdings}_i + \beta_5 \text{Pre-Issue Price}_i \\ &+ \beta_6 \text{NYSE}_i + \beta_7 \text{Public}_i + \beta_8 \Delta \text{PrePost}_i \\ &+ \sum_{t=1993\text{Jul}}^{2006\text{Apr}} \beta_{9t} \text{YearMonthDum}_{i,t} + \varepsilon_i. \end{aligned} \quad (1)$$

Explanatory variables are defined below. All change variables are calculated as deviations from control firms.

- ΔSI is change in short interest (number of shares) divided by total shares outstanding. The change in

short interest is the difference between short interest in the issue month and short interest in the previous month. This measure is a proxy for arbitrage activity.

- $\Delta \text{Market Cap}$ is the change in (log) market capitalization, defined as (log) average daily shares outstanding times closing stock price. The change in market capitalization is calculated as post-period market capitalization minus pre-period capitalization.
- $\Delta \text{Volatility}$ (used in liquidity regressions only) is the change in the standard deviation of daily returns.
- $\Delta \text{Institutional Holdings}$ is the change in institutional holdings (shares held by 13f institutions) divided by total shares outstanding. The change is calculated as the holdings at the issuing calendar year end minus the holdings at the prior calendar year end.
- *Pre-Issue Price* is the average (log) price during the pre-issue period.
- *NYSE* is a dummy variable, equal to one if the firm is listed on NYSE and zero otherwise.
- *Public* is a dummy variable, equal to one if the convertible bond is a public offering, and zero otherwise.
- $\Delta \text{PrePost}$ is the number of days between the pre- and the post-issue period.²⁵
- YearMonthDum_t are year and month fixed effects, indicating timing of the convertible bond issue.

The estimated coefficient on ΔSI is of primary interest. We expect to observe a positive role for ΔSI in liquidity changes (Prediction P1) and no impact of ΔSI on changes in price efficiency (Prediction P2). Control variables include changes in size, stock return volatility, and institutional holdings (a proxy for the supply of shares to short sellers). We also control for changes in volatility in the liquidity regressions due to relationships found in the literature. For example, Pastor and Stambaugh (2003) find correlation of 0.57 between market illiquidity and volatility. Spiegel and Wang (2005) report high correlation between idiosyncratic volatility and liquidity (i.e., liquidity produces perfect volatility sorts for a cross-section of stocks). We anticipate more room for marginal liquidity improvements in less-liquid stocks, so pre-issue price is included as a proxy for liquidity level (price and liquidity are negatively correlated; higher-priced stocks have lower spreads due to the fixed component). We also allow for variation based on exchange and whether an issue is public or private (e.g., liquidity of the bond issue might be higher for public issues), and we also include time effects.

The main results are presented in Table 4. All standard errors are heteroskedasticity-consistent and include industry clustering based on Fama and French (1997) industry definitions. The proxy for convertible bond arbitrage activity (ΔSI) is significantly and positively related to liquidity improvements based on five of the seven liquidity measures: number of trades, turnover,

²⁴ The inputs to the score measure for a year t issue are calculated as follows: turnover_{t-1} is the average monthly turnover in year $t - 1$; marketcap_{t-1} and $\text{book-to-market}_{t-1}$ are year-end values. The mean (median) score of the 846 control firms is 0.241 (0.198).

²⁵ In order to separate the potential impact of the announcement on liquidity and efficiency variables, we skip the 20 trading days prior to the announcement of the issue through 20 trading days following the issue date.

Table 4

Regression of liquidity and efficiency changes on ΔSI and firm characteristics.

This table shows results of a regression in which equity market liquidity and efficiency changes are regressed on issue period change in short interest (ΔSI , the proxy for convertible bond arbitrage activity) and firm characteristics. ΔSI is the change in short interest (from the month prior to issuance) divided by the number of shares outstanding in the month prior to issuance. The “pre-issue period” is the six months ending one month prior to announcement. The “post-issue period” is the six months beginning one month after issuance.

$\Delta Turnover$, $\Delta Trades$, and $\Delta Amihud$ are changes in the post-issue log *Turnover*, log *Number of trades*, and log *Amihud*, respectively, from the pre-issue measures. $\Delta OIBNUM$, $\Delta Spread$, $\Delta Depth$, and $\Delta Spr/dep$ are the percentage changes in the post-issue *OIBNUM*, *Percentage spread*, *Total depth/shares outstanding*, and *Dollar spread/dollar depth*, respectively, from the pre-issue measures. $\Delta |AR(1)|$ and $\Delta |VR(5) - 1|$ are the percentage changes in the post-issue $|Daily AR(1)|$ and $|Variance ratio(5) - 1|$, respectively, from the pre-issue measures.

Turnover is the average daily stock volume divided by shares outstanding. *Number of trades* is the average daily number of stock transactions. *Amihud* is the average daily ratio of absolute return to dollar volume (Amihud, 2002). *OIBNUM* is the average daily absolute difference between the numbers of buyer- and seller-initiated trades divided by their sum. *Percentage spread* is the difference between bid and ask quotes (time-weighted), expressed as percentage of bid–ask midpoint. *Total depth/shares outstanding* is the sum of bid and ask quoted depths divided by shares outstanding. *Dollar spread/dollar depth* is the ratio of quoted spread to quoted depth, both expressed in dollars. $|Daily AR(1)|$ is the absolute value of first-order autocorrelation of daily returns. $|Variance ratio(5) - 1|$ is the absolute deviation of five-day variance ratio from one.

$\Delta Market cap$ and $\Delta Volatility$ are the post-issue equity market capitalization and standard deviation of daily return minus the corresponding pre-issue period measures. $\Delta Institutional holdings$ is the change in institutional holdings (by 13f institutions) divided by shares outstanding at issuing calendar year end from prior calendar year end. *Pre-issue price* is the average (log) price in pre-issue period. *NYSE* and *Public* are dummy variables. $\Delta Pre post$ is the number of days between the last day in pre-issue period and the first day in post-issue period. Estimates on the year-month dummies are not reported.

All change variables are expressed as deviation from control firms (issuer minus control). Control firms are matched based on size, book-to-market, and turnover before issuance, exchange, and industry. Heteroskedasticity-consistent industry-clustered *t*-statistics are in parentheses. *, **, and *** denote 10%, 5%, and 1% significance, respectively. The sample period is from July 1993 to May 2006.

	Liquidity measures						Efficiency measures											
	Δ Turnover	<i>t</i> -stat	Δ Trades	<i>t</i> -stat	Δ Amihud	<i>t</i> -stat	Δ OIBNUM	<i>t</i> -stat	Δ Spread	<i>t</i> -stat	Δ Depth	<i>t</i> -stat	Δ Spr/dep	<i>t</i> -stat	Δ $ AR(1) $	<i>t</i> -stat	Δ $ VR(5) - 1 $	<i>t</i> -stat
Intercept	0.091	(0.58)	−0.116	(−1.23)	−0.192	(−0.65)	0.064	(0.59)	0.030	(0.36)	0.648***	(3.31)	−0.277	(−1.39)	1.947	(0.25)	4.948	(1.04)
ΔSI	2.993***	(3.16)	2.474***	(4.23)	−3.042***	(−3.75)	−0.460	(−1.01)	0.375	(0.85)	1.950***	(2.69)	−1.656**	(−2.23)	−46.239	(−0.71)	−3.528	(−0.17)
$\Delta Market cap$	0.170***	(4.97)	0.617***	(19.39)	−0.982***	(−21.84)	−0.192***	(−9.92)	−0.512***	(−12.73)	−0.733***	(−6.53)	−0.209***	(−4.41)	−5.828	(−0.62)	−1.639**	(−2.00)
$\Delta Volatility$	10.063***	(6.47)	9.552***	(6.63)	−0.369	(−0.36)	−2.294***	(−2.75)	2.944***	(2.70)	5.976***	(2.69)	−0.064	(−0.09)				
$\Delta Institutional holdings$	0.438***	(8.62)	0.158***	(3.00)	−0.468***	(−4.15)	−0.038	(−1.07)	−0.213***	(−3.69)	−0.116	(−0.95)	−0.113	(−1.52)	0.811	(0.10)	−0.812	(−0.56)
Pre-issue price $\times$ 100	−0.260	(−0.09)	3.976**	(2.07)	1.125	(0.34)	−1.931*	(−1.90)	0.325	(0.17)	−9.983***	(−2.81)	−5.716	(−1.01)	117.303	(0.70)	−32.742	(−0.36)
NYSE	−0.080**	(−2.27)	−0.119***	(−4.22)	−0.020	(−0.40)	0.048**	(2.02)	−0.016	(−0.84)	0.084	(0.94)	−0.006	(−0.11)	4.034	(1.08)	−0.647	(−0.51)
Public	−0.027	(−0.52)	0.058	(1.19)	0.099*	(1.71)	−0.037	(−1.55)	0.059*	(1.81)	−0.094	(−0.99)	0.045	(0.67)	−16.484	(−1.11)	3.549	(1.44)
$\Delta Pre post \times 1000$	1.466*	(1.85)	0.330	(0.60)	−0.685	(−0.50)	−0.581	(−1.06)	−0.608	(−1.09)	−0.581	(−0.35)	2.874	(0.97)	−14.450	(−0.11)	−24.060	(−1.08)
Number of observations	846		846		846		846		846		846		846		846		846	
Adjusted R^2 (%)	36.81		56.36		57.81		33.42		58.03		35.95		19.87		10.47		59.58	

Amihud, depth, and *spread/depth*. We do not observe systematic variation of ΔSI with either order imbalance or spread. The spread result is somewhat puzzling because this measure showed significant changes based on portfolio sorts; however, change in market capitalization already captures a price change, and therefore the coefficient on percentage spread (change in spread divided by price) in the regression equation may pick up changes in dollar spread, rather than percentage spread. Importantly, the *spread/depth* measure, a more complete measure of the quality of the quote than spread or depth alone, suggests a significant relationship between convertible bond arbitrage activity and liquidity improvements. Although it is not the main focus of the analysis, it is also interesting to note that liquidity increases with increases in institutional holdings, market capitalization, and volatility. Neither of the price-efficiency measures are related to ΔSI . We interpret this as evidence that these traders do not enhance efficiency and simply provide supply to equity markets, as in the prediction (P2).²⁶ In the next section, we provide another returns-based efficiency test.

An assumption underlying much of the analysis in this paper is that the change in short interest near issuance is due to convertible bond arbitrage activity, which we do not directly observe. It is possible that the observed short selling near issuance is being driven by valuation shorting. We attempt to separate other possible explanations of the observed short selling by conducting several tests of whether the data are consistent with convertible bond arbitrage. First, we examine long-run stock returns following issuance. If the increase in short interest is due to valuation shorting, we would expect short sellers to make money from their positions. We then conduct a second test using hand-collected announcement dates for the sample of issuers to isolate the impact of convertible bond announcement- versus issue-period shorting, which we interpret as valuation shorting and convertible bond arbitrage, respectively. The basic intuition is that valuation short sellers will be able to trade at the announcement of the bond, whereas convertible bond arbitrageurs will not short until the actual issue of the bond. This is done in order to avoid stock exposure that would result if convertible bond arbitrageurs took short equity positions

prior to issuance. The results of these tests are consistent with short selling due to convertible bond arbitrage and are presented below.

5.3. Long-run returns

As a final efficiency check, we test whether short sellers make money from their equity market positions. The primary goal is to shed further light on the possibilities noted in Section 3, that issue-period short sellers are either: (1) privately informed about the value of the underlying equity or (2) that they exploit known price inefficiencies such as return autocorrelation. Evidence of negative long-run returns following issuance would be consistent with these possibilities. A finding of no abnormal returns following issuance would be consistent with prediction P2, that convertible bond arbitrageurs do not play a role in stock price efficiency.

Results are presented in Table 5 and are consistent with the findings for the other efficiency measures: there is no evidence that issue-period short sellers make money from their positions; however, we do observe a significantly negative return of -1.30% per day between announcement and issue date (when they are at least two trading days apart). Therefore, it may be that valuation shorts make money based on positions taken at the announcement of the issue (Day X in Fig. 4), while short sellers engaged in convertible bond arbitrage strategies take positions near issuance (Day 0 in Fig. 4) and do not earn abnormal returns from equity positions. Importantly, this also implies that there is no evidence that valuation shorts will have incentive to maintain their short positions following issuance and are unlikely to impact post-issuance liquidity changes that we observe in the main analysis. This is because abnormal returns from the short positions occur only between announcement and issuance.

Table 5 also suggests that there may be important differences between short selling observed on announcement versus issue dates. These differences might provide powerful identification for the type of short selling (i.e., convertible bond arbitrage versus other types of arbitrage) that we are observing in the data. In the next section, we delve deeper into this question by using the data on announcement dates. This allows us to separate the impact of short selling due to the market's knowledge that the bond is being issued from actual convertible bond arbitrage (which would be more likely to coincide with the issue itself).

5.4. Robustness analysis: convertible bond arbitrage versus valuation shorts

Because we do not observe convertible bond arbitrage activity directly, we face the challenge that we may be picking up the impact of short selling on liquidity and efficiency due to some other factor. A particular concern is valuation short selling due to news that the firm is raising convertible debt. In order to address this issue, we hand-collect announcement dates of all convertible bond issues

²⁶ In addition to the control firms identified according to the procedure outlined in Section 5.2.2, we conduct robustness checks based on two “issue matches.” We match convertible bond issuers to a sample of straight-debt issuers (matched by exchange, industry, size, book-to-market, and issue size). The basic aim is to distinguish the effect of convertible bond issuance from a general increase in leverage. Results are similar to those in Table 4: we find that ΔSI is positively and significantly related to increases in turnover and number of trades; and ΔSI is negatively and significantly related to both spread and Amihud (2002) illiquidity measure. In our second “issue match,” we match convertible bond issuers to firms issuing seasoned equity because purchasers of the equity issue would not need to manage a short inventory, as is the case for convertible bond arbitrage. Results are similar to those in the straight-debt analysis. In general, we find that the impact of ΔSI is somewhat stronger in these robustness checks than in the main analysis. Detailed results and the matching procedure are available upon request. We thank William Fung for encouraging the equity issuer robustness check.

Table 5

Long-run returns.

This table summarizes post-issuance and post-announcement returns over 6-, 12-, 18-, and 24-month horizons. In Panels A and B, *Cumulative return* is the average holding period return from long stock positions starting after market close on the convertible bond issue date and announcement date, respectively. Panel C shows the average daily return of issuers and control firms between market close on announcement date and the trading day prior to issue date, for issues that are announced at least two trading days prior to issuance. Control firms are matched based on size, book-to-market, and turnover before issuance, exchange, and industry. Industry- and time-clustered *t*-statistics of the differences in returns are in parentheses. *** denotes 1% significance. The sample period is from July 1993 to May 2006.

Number of observations = 846.

Cumulative return (not annualized)	Issue date	+6 months	+12 months	+18 months	+24 months
<i>Panel A: Long-run returns after issuance</i>					
Issuing firm		6.57%	9.22%	12.24%	16.53%
Control firm		5.47%	8.85%	13.68%	17.66%
Difference		1.10%	0.37%	−1.44%	−1.13%
<i>t</i> -stat		(0.58)	(0.13)	(−0.41)	(−0.31)
Cumulative return (not annualized)	Announcement date	+6 months	+12 months	+18 months	+24 months
<i>Panel B: Long-run returns after announcement</i>					
Issuing firm		3.94%	6.64%	9.73%	13.82%
Control firm		6.26%	10.16%	14.00%	18.31%
Difference		−2.32%	−3.52%	−4.27%	−4.49%
<i>t</i> -stat		(−1.14)	(−1.14)	(−1.21)	(−1.21)
Number of observations = 348.					
<i>Panel C: Returns between announcement and issuance</i>					
Average daily return between announcement and issue dates					
Issuing firm			−1.30%		
Control firm			0.13%		
Difference			−1.43%***		
<i>t</i> -stat			(−6.80)		
Average number of trading days			10.21		

in the sample.²⁷ We identify 132 issues for which the announcement date and the issue date are such that the observed change in short interest for announcement and issuance occur during different short-interest reporting months. This allows us to clearly separate valuation shorting from convertible bond arbitrage. Table 6 presents the results of analysis for this subsample. The findings are consistent with the main regressions in Table 4. With the exception of depth and spread-to-depth, all liquidity changes that are significantly related to the proxy for convertible bond arbitrage in the main analysis are also significant in this subsample analysis. Most important, changes in short interest during the announcement month ($\Delta SI_{\text{Announcement}}$) are not significantly related to any observed liquidity and efficiency changes.

5.5. Robustness analysis: potential endogeneity of arbitrage

It is possible that convertible bond arbitrageurs are attracted to stocks for which they expect increases in liquidity. While our prior is that the direction of the causality runs from the arbitrageurs, we explicitly account for potential simultaneity by estimating a simultaneous

equations model of both changes in market quality and short-interest changes. We estimate the following system:

$$\begin{aligned} \Delta Liquidity_i &= \alpha + \beta_1 \Delta SI_i^{IV} + \beta_2 \Delta Market\ Cap_i + \beta_3 \Delta Volatility_i \\ &+ \beta_4 \Delta Institutional\ Holdings_i + \beta_5 Pre\text{-}Issue\ Price_i + \beta_6 NYSE_i \\ &+ \beta_7 Public_i + \beta_8 \Delta PrePost_i + \sum_{t=1993Jul}^{2006Apr} \beta_{9t} YearMonthDum_{i,t} + \varepsilon_i \end{aligned} \quad (2)$$

and

$$\begin{aligned} \Delta SI_i &= \alpha + \beta_1 \Delta Liquidity_i^{IV} + \beta_2 Dollar\ Volume_i + \beta_3 Volatility_i \\ &+ \beta_4 Institutional\ Holdings_i + \beta_5 Dividends_i \\ &+ \beta_6 Conversion\ Premium_i \\ &+ \beta_7 NYSE_i + \beta_8 Public_i + \beta_9 \Delta PrePost_i \\ &+ \sum_{t=1993Jul}^{2006Apr} \beta_{10t} YearMonthDum_{i,t} + \varepsilon_i. \end{aligned} \quad (3)$$

As in Table 4, all change variables are calculated as deviations from firm *i*'s control firm in order to control for market-wide trends in the variables of interest. The explanatory variables in the liquidity change regressions are identical to those in Table 4, except ΔSI_i^{IV} is instrumented. Similarly, liquidity change variables are

²⁷ We use Lexis-Nexis and Factiva news searches to identify the earliest date on which the bond issue is mentioned in the news.

Table 6

Identification of convertible bond arbitrageurs versus valuation shorts: analysis of the impact of shorting at announcement compared to issue date.

This table shows results of a regression in which equity market liquidity and efficiency changes are regressed on change in short interest (ΔSI and $\Delta SI_Announcement$, calculated at issuance and announcement, respectively) and firm characteristics. ΔSI is the proxy for convertible bond arbitrage activity and is defined as the change in short interest (from the month prior to issuance) divided by the number of shares outstanding in the month prior to issuance. $\Delta SI_Announcement$ is calculated in the same way but is based on announcement month rather than issue month. Only issues where issuing and announcement months fall into different monthly short-interest files are included. The “pre-issue period” is the six months ending one month prior to announcement. The “post-issue period” is the six months beginning one month after issuance.

$\Delta Turnover$, $\Delta Trades$, and $\Delta Amihud$ are changes in the post-issue log *Turnover*, log *Number of trades*, and log *Amihud*, respectively, from the pre-issue measures. $\Delta OIBNUM$, $\Delta Spread$, $\Delta Depth$, and $\Delta Spr/dep$ are the percentage changes in the post-issue *OIBNUM*, *Percentage spread*, *Total depth/shares outstanding*, and *Dollar spread/dollar depth*, respectively, from the pre-issue measures. $\Delta |AR(1)|$ and $\Delta |VR(5) - 1|$ are the percentage changes in the post-issue $|Daily AR(1)|$ and $|Variance ratio(5) - 1|$, respectively, from the pre-issue measures.

Turnover is the average daily stock volume divided by shares outstanding. *Number of trades* is the average daily number of stock transactions. *Amihud* is the average daily ratio of absolute return to dollar volume (Amihud, 2002). *OIBNUM* is the average daily absolute difference between the numbers of buyer- and seller-initiated trades divided by their sum. *Percentage spread* is the difference between bid and ask quotes (time-weighted), expressed as percentage of bid–ask midpoint. *Total depth/shares outstanding* is the sum of bid and ask quoted depths divided by shares outstanding. *Dollar spread/dollar depth* is the ratio of quoted spread to quoted depth, both expressed in dollars. $|Daily AR(1)|$ is the absolute value of first-order autocorrelation of daily returns. $|Variance ratio(5) - 1|$ is the absolute deviation of five-day variance ratio from one.

$\Delta Market cap$ and $\Delta Volatility$ are the post-issue equity market capitalization and standard deviation of daily return minus the corresponding pre-issue period measures. $\Delta Institutional holdings$ is the change in institutional holdings (by 13f institutions) divided by shares outstanding at issuing calendar year end from prior calendar year end. *Pre-issue price* is the average (log) price in pre-issue period. *NYSE* and *Public* are dummy variables. $\Delta Pre post$ is the number of days between the last day in pre-issue period and the first day in post-issue period. Estimates on the year-month dummies are not reported.

All change variables are expressed as deviation from control firms (issuer minus control). Control firms are matched based on size, book-to-market, and turnover before issuance, exchange, and industry. Heteroskedasticity-consistent industry-clustered *t*-statistics are in parentheses. *, **, and *** denote 10%, 5%, and 1% significance, respectively. The sample period is from July 1993 to May 2006.

	Liquidity measures								Efficiency measures									
	Δ Turnover	t -stat	Δ Trades	t -stat	Δ Amihud	t -stat	Δ OIBNUM	t -stat	Δ Spread	t -stat	Δ Depth	t -stat	Δ Spr/dep	t -stat	Δ AR(1)	t -stat	Δ VR(5) − 1	t -stat
Intercept	−0.414	(−0.81)	−0.609	(−1.35)	0.793	(1.06)	0.208	(0.93)	0.053	(0.20)	0.625	(1.65)	0.887	(1.61)	7.563	(0.52)	0.064	(0.00)
ΔSI	9.680**	(2.43)	8.345**	(2.25)	−11.594**	(−2.59)	−1.714	(−0.98)	−1.319	(−0.94)	6.022	(1.34)	−5.984	(−1.57)	−105.969	(−0.96)	−41.277	(−0.38)
$\Delta SI_Announcement$	1.446	(0.24)	−5.137	(−0.64)	8.512	(0.89)	−0.501	(−0.17)	3.914	(1.15)	−4.645	(−0.56)	7.020	(1.65)	−185.934	(−1.00)	−134.303	(−0.88)
$\Delta Market\ cap$	0.060	(0.29)	0.428**	(2.30)	−0.770***	(−3.07)	−0.177*	(−1.68)	−0.511***	(−5.45)	−0.601***	(−2.91)	−0.092	(−0.32)	−0.975	(−0.17)	−3.814	(−0.57)
$\Delta Volatility$	8.027***	(3.37)	5.105*	(1.96)	−4.295	(−1.30)	−0.779	(−0.38)	−2.003**	(−2.26)	5.175*	(1.77)	−3.644**	(−2.29)				
$\Delta Institutional\ holdings$	0.485**	(2.30)	0.093	(0.34)	−0.185	(−0.54)	−0.131	(−0.78)	−0.385***	(−3.41)	−0.366	(−1.31)	−0.159	(−0.77)	5.621	(0.57)	0.529	(0.10)
Pre-issue price × 100	15.117	(0.98)	20.535	(1.63)	−27.128	(−1.41)	−1.689	(−0.28)	−1.174	(−0.22)	−3.577	(−0.28)	−31.369*	(−1.93)	−40.640	(−0.12)	98.382	(0.33)
NYSE	−0.097	(−0.58)	−0.114	(−0.92)	0.228	(0.77)	−0.136	(−1.20)	0.006	(0.06)	−0.253	(−1.22)	0.192	(1.13)	−4.848	(−1.08)	4.654	(0.75)
Public	−0.080	(−0.36)	0.022	(0.10)	0.073	(0.24)	−0.076	(−0.90)	0.111*	(1.74)	−0.001	(−0.01)	0.185	(1.14)	−10.027	(−1.27)	8.833	(1.01)
$\Delta Pre\ post \times 1000$	−1.487	(−0.95)	−2.150	(−1.32)	0.910	(0.24)	−0.168	(−0.08)	−0.052	(−0.07)	−4.122	(−1.12)	2.455	(1.05)	32.060	(0.54)	−36.330	(−0.48)
Number of observations	132		132		132		132		132		132		132		132		132	
Adjusted R^2 (%)	75.70		81.64		68.40		72.51		89.10		70.80		91.43		70.29		78.67	

instrumented in the ΔSI_i regressions. The other explanatory variables in the ΔSI_i regression are chosen to proxy for characteristics of firms that tend to be attractive to convertible bond arbitrageurs²⁸.

- *Dollar Volume* is the mean daily dollar volume during the pre-issue period. This variable is included to capture the impact of stock liquidity levels on convertible bond arbitrage activity. It is easier to dynamically hedge more liquid stocks.
- *Volatility* is the mean standard deviation of daily returns during the pre-issue period. Convertible bond arbitrageurs are expected to prefer higher volatility issuers. This is because there are higher potential trading profits due to the embedded option in the bond.
- *Institutional Holdings* is the level of institutional holdings (shares held by 13f institutions) divided by shares outstanding at calendar year end prior to issuance. This variable is a proxy for the availability of shares to borrow.
- *Dividends* are stock dividend yields and are included because convertible bond arbitrageurs are expected to prefer low or no dividend-paying stocks since short sellers have to pay dividends.
- *Conversion Premium* is the amount (in percent) by which the conversion price exceeds the market value of the common stock at issuance. Calamos (2003, p. 25) states that arbitrageurs tend to prefer stocks with conversion premia that are less than 25% because a low conversion premium implies lower interest rate and credit risk.
- *NYSE* is a dummy variable, equal to one if the firm is listed on NYSE and zero otherwise.
- *Public* is a dummy variable, equal to one if the convertible bond is a public offering, and zero otherwise.
- $\Delta PrePost$ is the number of days between the pre- and the post-issue period.
- $YearMonthDum_i$ are year and month fixed effects, indicating timing of the convertible bond issue.

In the first-stage regressions for arbitrage activity, we use percentage of shares affected by the issue (conversion ratio * number of bonds/shares outstanding) and analyst opinion prior to issuance (percentage of buy recommendations) as instruments, in addition to the explanatory variables specified in the simultaneous equations. These instruments are chosen because convertible bond arbitrageurs will short more when a high percentage of shares are affected by the issue. Analyst recommendations are included because short sellers can have trading strategies based on these recommendations. In the first-stage regressions for $\Delta Liquidity_i$, we include change in analyst coverage and change in absolute price deviation from \$30 as instruments (in addition to the other exogenous variables). We expect that there is a high correlation

between analyst coverage and liquidity levels, and that a stock price close to \$30 indicates higher liquidity.²⁹

The results from simultaneous equations for each liquidity measure are presented in Table 7. The main findings are qualitatively similar to the main regression (with the exception of depth, for which we do not observe a statistically significant increase after controlling for potential endogeneity). Moreover, the liquidity changes do not impact arbitrage activity, which is proxied by ΔSI (Panel B of the table).

5.6. Robustness analysis: other potential sources of volume

It is possible that following the issue of a convertible bond, there is a mechanical increase in trading volume due to a potential future increase in shares outstanding (i.e., investors eventually exercising the conversion option).³⁰ In this case, we would expect to observe increases in volume and turnover that are proportional to the fraction of new shares. We may also observe corresponding increases in other liquidity measures. We would like to distinguish increases in volume from hedging trades due to convertible bond arbitrage (i.e., the post-issuance trading activity depicted in Fig. 1) and increases in volume due to other sources.³¹ To address the important concern that the results may be driven by other sources of volume, we conduct two sets of robustness tests.

First, we replicate the main analysis (i.e., Table 4 regressions), but add an explicit control for expected increases in volume (unrelated to convertible bond arbitrage activity), which is proportional to the fraction of new shares:

$$E(\Delta Volume) = \left[\frac{\text{shares underlying the issue}}{\text{shares outstanding prior to issue}} * \text{pre-issue}(\log) \text{volume} \right].^{32}$$

The results are presented in Table 8 and show that expected changes in volume are positively and significantly related to realized changes for trading activity measures that are linked to volume (turnover, number of trades, and the Amihud, 2002, illiquidity measure), as one

²⁹ The percentage of shares affected is highly significant (*t*-statistic of 6.21) in the first-stage regression of arbitrage activity. We thank an anonymous referee for suggesting the importance of the conversion ratio. Analyst opinion is also significant (*t*-statistic of 1.92) in the first-stage. For first-stage liquidity regressions, the analyst coverage variable is significant for four of seven measures: turnover, number of trades, order imbalance, and depth. Price deviation from \$30 is significant in spread and depth. We use all exogenous variables and instruments in the first-stage regressions. We also confirm that the instruments for the liquidity changes are not significant in the ΔSI regressions and vice versa. First-stage results are not reported (for brevity) and are available upon request.

³⁰ For example, Stein (1992) shows that convertible bonds can essentially be a “backdoor” to equity financing. It is also possible that the introduction of the new security causes overall volume to increase; however, this would be picked up by the estimated intercepts (as in Kumar, Sarin, and Shastri, 1998), not the coefficient on ΔSI , the proxy for arbitrage activity.

³¹ We thank an anonymous referee for suggesting this distinction.

³² Shares underlying the issue is defined as the conversion shares ratio (reported in SDC Platinum) times the number of bonds.

²⁸ See Calamos (2003, p. 25).

Table 7

Simultaneous equations estimation: change in liquidity and ΔSI .

This table presents a simultaneous equation system of equity market liquidity changes and issue period change in short interest (ΔSI), the proxy for convertible bond arbitrage activity. ΔSI is the change in short interest (from the month prior to issuance) divided by the number of shares outstanding in the month prior to issuance. Panel A shows the regressions of liquidity changes on ΔSI^{IV} and other variables, while Panel B shows the regressions of ΔSI on $\Delta Liquidity^{IV}$ and other variables. First-stage results are not reported. The “pre-issue period” is the six months ending one month prior to announcement. The “post-issue period” is the six months beginning one month after issuance.

$\Delta Turnover$, $\Delta Trades$, and $\Delta Amihud$ are changes in the post-issue log *Turnover*, log *Number of trades*, and log *Amihud*, respectively, from the pre-issue measures. $\Delta OIBNUM$, $\Delta Spread$, $\Delta Depth$, and $\Delta Spr/dep$ are the percentage changes in the post-issue *OIBNUM*, *Percentage spread*, *Total depth/shares outstanding*, and *Dollar spread/dollar depth*, respectively, from the pre-issue measures. $\Delta |AR(1)|$ and $\Delta |VR(5) - 1|$ are the percentage changes in the post-issue $|Daily\ AR(1)|$ and $|Variance\ ratio(5) - 1|$, respectively, from the pre-issue measures.

Turnover is the average daily stock volume divided by shares outstanding. *Number of trades* is the average daily number of stock transactions. *Amihud* is the average daily ratio of absolute return to dollar volume (Amihud, 2002). *OIBNUM* is the average daily absolute difference between the numbers of buyer- and seller-initiated trades divided by their sum. *Percentage spread* is the difference between bid and ask quotes (time-weighted), expressed as percentage of bid–ask midpoint. *Total depth/shares outstanding* is the sum of bid and ask quoted depths divided by shares outstanding. *Dollar spread/dollar depth* is the ratio of quoted spread to quoted depth, both expressed in dollars. $|Daily\ AR(1)|$ is the absolute value of first-order autocorrelation of daily returns. $|Variance\ ratio(5) - 1|$ is the absolute deviation of five-day variance ratio from one.

In Panel A, ΔSI^{IV} is estimated from a first-stage regression (see Section 5.5 for specification). $\Delta Market\ cap$ and $\Delta Volatility$ are the post-issue equity market capitalization and standard deviation of daily return minus the corresponding pre-issue period measures. $\Delta Institutional\ holdings$ is the change in institutional holdings (by 13f institutions) divided by shares outstanding at issuing calendar year end from prior calendar year end. *Pre-issue price* is the average (log) price in pre-issue period. *NYSE* and *Public* are dummy variables. $\Delta Pre\ post$ is the number of days between the last day in pre-issue period and the first day in post-issue period. Estimates on the year-month dummies are not reported.

In Panel B, $\Delta Liquidity^{IV}$ are the different liquidity changes ($\Delta Turnover$, $\Delta Trades$, $\Delta Amihud$, $\Delta OIBNUM$, $\Delta Spread$, $\Delta Depth$, and $\Delta Spr/dep$) estimated from first-stage regressions (see Section 5.5 for specification). *Pre-issue dollar volume* and *Pre-issue volatility* are, respectively, the average (log) dollar volume and standard deviation of daily return in pre-issue period. *Pre-issue institutional holdings* are the institutional holdings divided by shares outstanding at calendar year end prior to issuance. *Pre-issue dividend* is the dividend yield one year prior to issuance. *Conversion premium* is the amount (in percent) by which the conversion price exceeds the market value of the common stock at issuance. Estimates on the year-month dummies are not reported.

All change variables are expressed as deviation from control firms (issuer minus control). Control firms are matched based on size, book-to-market, and turnover before issuance, exchange, and industry. Heteroskedasticity-consistent industry-clustered *t*-statistics are in parentheses. *, **, and *** denote 10%, 5%, and 1% significance, respectively. The sample period is from July 1993 to May 2006.

Panel A: Liquidity changes														
	Δ Turnover	<i>t</i> -stat	Δ Trades	<i>t</i> -stat	Δ Amihud	<i>t</i> -stat	Δ OIBNUM	<i>t</i> -stat	Δ Spread	<i>t</i> -stat	Δ Depth	<i>t</i> -stat	Δ Spr/dep	<i>t</i> -stat
Intercept	−0.124	(−0.72)	−0.237**	(−2.18)	−0.055	(−0.21)	0.076	(0.67)	0.004	(0.05)	0.569***	(2.61)	−0.187	(−1.00)
ΔSI^{IV}	10.963***	(4.43)	6.931***	(3.81)	−8.073***	(−3.19)	−0.888	(−1.06)	1.318	(1.63)	4.857	(1.38)	−4.971*	(−1.94)
$\Delta Market\ cap$	0.142***	(2.72)	0.601***	(16.98)	−0.964***	(−18.44)	−0.190***	(−10.07)	−0.516***	(−12.49)	−0.743***	(−5.94)	−0.197***	(−4.02)
$\Delta Volatility$	10.487***	(7.64)	9.789***	(7.42)	−0.637	(−0.54)	−2.317***	(−2.85)	2.994***	(2.89)	6.131***	(2.74)	−0.240	(−0.29)
$\Delta Institutional\ holdings$	0.408***	(7.11)	0.141**	(2.48)	−0.449***	(−3.84)	−0.037	(−1.00)	−0.217***	(−3.77)	−0.127	(−0.99)	−0.101	(−1.31)
Pre-issue price × 100	4.646	(1.32)	6.719***	(2.79)	−1.971	(−0.66)	−2.194**	(−2.01)	0.905	(0.48)	−8.193**	(−2.32)	−7.757	(−1.18)
NYSE	−0.027	(−0.45)	−0.089**	(−2.23)	−0.054	(−0.84)	0.045*	(1.92)	−0.010	(−0.53)	0.104	(0.96)	−0.028	(−0.52)
Public	0.028	(0.55)	0.089*	(1.79)	0.064	(1.04)	−0.040	(−1.55)	0.065**	(2.08)	−0.074	(−0.85)	0.022	(0.31)
$\Delta Pre\ post \times 1000$	0.233	(0.28)	−0.360	(−0.50)	0.093	(0.00)	−0.510	(−0.99)	−0.750	(−1.34)	−1.030	(−0.60)	3.386	(1.06)
Number of observations	846		846		846		846		846		846		846	
Adjusted R^2 (%)	18.94		44.25		46.81		18.80		48.68		21.63		2.08	

Table 7 (continued)

All coefficients are $\times 1000$																					
Panel B: ΔSI																					
When $\Delta Liquidity^{IV} =$																					
	Δ	Turnover	t-stat	Δ	Trades	t-stat	Δ	Amihud	t-stat	Δ	OIBNUM	t-stat	Δ	Spread	t-stat	Δ	Depth	t-stat	Δ	Spr/dep	t-stat
Intercept	11.448*	(1.78)	12.987*	(1.85)	11.477*	(1.83)	13.528*	(1.90)	13.423*	(1.67)	15.181*	(1.69)	7.631	(1.02)							
$\Delta Liquidity^{IV}$	9.686	(0.89)	5.302	(0.71)	-5.270	(-1.47)	-17.790	(-0.77)	-9.180	(-1.55)	-3.730	(-0.99)	-11.740	(-0.64)							
Pre-issue dollar volume	-0.710	(-0.60)	-1.120	(-1.01)	-1.250	(-1.07)	-1.210	(-1.08)	-1.520	(-1.15)	-1.410	(-1.07)	-1.360	(-1.14)							
Pre-issue volatility	199.239*	(1.95)	175.534**	(1.96)	142.304***	(2.70)	188.692*	(1.73)	146.362**	(2.50)	162.161**	(2.37)	134.291***	(2.73)							
Pre-issue institutional holdings	6.785**	(2.15)	5.471*	(1.71)	5.937*	(1.92)	5.227	(1.52)	5.440*	(1.72)	5.493	(1.60)	5.434*	(1.70)							
Pre-issue dividend	-7.350	(-1.34)	-7.040	(-1.20)	-7.640	(-1.35)	-7.760	(-1.11)	-7.010	(-1.22)	-5.870	(-1.09)	-11.330	(-0.99)							
Conversion premium (%)	-0.050**	(-2.42)	-0.050***	(-2.57)	-0.040*	(-1.88)	-0.060***	(-2.91)	-0.050**	(-2.29)	-0.060***	(-3.24)	-0.040	(-1.15)							
NYSE	-7.120*	(-1.82)	-7.400**	(-2.04)	-7.860**	(-2.48)	-7.300**	(-1.96)	-8.180**	(-2.56)	-8.040**	(-2.69)	-8.200**	(-2.56)							
Public	-6.510**	(-1.96)	-7.460**	(-2.36)	-6.580**	(-2.07)	-7.950**	(-2.21)	-6.790**	(-2.20)	-7.690**	(-2.38)	-6.430*	(-1.85)							
Δ Pre post	0.158***	(2.95)	0.176***	(2.97)	0.176***	(2.98)	0.169***	(3.23)	0.175***	(2.92)	0.179***	(3.06)	0.210**	(2.19)							
Number of observations	846		846		846		846		846		846		846								
Adjusted R^2 (%)	5.13		4.98		5.51		4.89		4.72		4.06		3.97								

might expect. However, these expected mechanical volume changes do not explain non-volume-based liquidity improvements (i.e., insignificant relationships with depth, order imbalance, and spread-to-depth, with the exception of a significant and *positive* relationship with percentage spread). More importantly for this analysis, the significant relationship between the arbitrage activity proxy (ΔSI) and all liquidity changes reported in the main analysis are largely unaffected by the inclusion of expected changes in volume.³³

A benefit of the measures above is that both the proxy for convertible arbitrage activity and expected changes in volume are calculated at time 0 relative to the issue date. This is a clean, well-defined window for measuring the presence of convertible bond arbitrageurs. Ideally, we would want to measure the impact of *actual* convertible bond arbitrage activity, controlling for *actual* changes in volume. A challenge in this paper's framework is that changes in actual volume include two components: changes in volume due to convertible bond arbitrage and non-arbitrage related changes (the mechanical changes described above or other sources). One possibility would be to put actual volume changes, measured over months +1 through +6, into the regression in which the convertible bond arbitrage activity proxy, ΔSI , is calculated at issue month 0. This approach is not appealing since one would expect actual volume changes to have the most power in explaining liquidity changes (simply because the volume change measure is calculated over the same +1 through +6 interval as the liquidity changes). In this case, it would be very difficult to distinguish whether the convertible bond arbitrage or non-arbitrage component of volume drives the change.

To circumvent this potential issue, we return to the intuition that convertible bond arbitrageurs will dynamically hedge following issuance: they will short more when stock prices rise and cover their short positions as stock prices fall. This will increase the volatility of short interest relative to the pre-issue period. In the second robustness test, we introduce an alternative proxy for dynamic hedging activity, *Dynamic_Hedge*, which is defined as the average (log) absolute value of short-interest changes during post-issue period minus the average absolute value of short-interest changes during the pre-issue period.³⁴ In Fig. 4, these are months +1 through +6 and X-6 and X-1, respectively. The intuition for *Dynamic_Hedge* is that monthly decreases

³³ In addition to expected increase in volume, we replicate the analysis using expected increase in turnover and dollar volume (since volume can be affected by changes in the number of shares outstanding). We also eliminate observations in which there is a change in the number of shares outstanding during the horizon of interest. All results are qualitatively similar to those shown in Table 8.

³⁴ In unreported analysis, we also conduct a third test, which controls for mechanical increases in volume due to the potential increase in shares outstanding. We match convertible bond issuers to a sample of straight equity issuers (matched by exchange, industry, size, book-to-market, and issue size), in which there is a clear increase in the number of shares outstanding. We re-run the analysis, where all variables are measured as deviations from this "issue-control" sample. As noted earlier, results are similar to those in the main analysis.

Table 8

Regression of liquidity and efficiency changes on ΔSI and expected volume increase.

This table presents results of a regression in which equity market liquidity and efficiency changes are regressed on issue period change in short interest (ΔSI , the proxy for convertible bond arbitrage activity), expected change in volume ($E(\Delta Volume)$), and firm characteristics. ΔSI is the change in short interest (from the month prior to issuance) divided by the number of shares outstanding in the month prior to issuance. $E(\Delta Volume)$ is the pre-issue average (log) daily volume in number of shares, times shares underlying the issue divided by shares outstanding prior to issuance. The “pre-issue period” is the six months ending one month prior to announcement. The “post-issue period” is the six months beginning one month after issuance.

$\Delta Turnover$, $\Delta Trades$, and $\Delta Amihud$ are changes in the post-issue log *Turnover*, log *Number of trades*, and log *Amihud*, respectively, from the pre-issue measures. $\Delta OIBNUM$, $\Delta Spread$, $\Delta Depth$, and $\Delta Spr/dep$ are the percentage changes in the post-issue *OIBNUM*, *Percentage spread*, *Total depth/shares outstanding*, and *Dollar spread/dollar depth*, respectively, from the pre-issue measures. $\Delta |AR(1)|$ and $\Delta |VR(5) - 1|$ are the percentage changes in the post-issue $|Daily\ AR(1)|$ and $|Variance\ ratio(5) - 1|$, respectively, from the pre-issue measures.

Turnover is the average daily stock volume divided by shares outstanding. *Number of trades* is the average daily number of stock transactions. *Amihud* is the average daily ratio of absolute return to dollar volume (Amihud, 2002). *OIBNUM* is the average daily absolute difference between the numbers of buyer- and seller-initiated trades divided by their sum. *Percentage spread* is the difference between bid and ask quotes (time-weighted), expressed as percentage of bid–ask midpoint. *Total depth/shares outstanding* is the sum of bid and ask quoted depths divided by shares outstanding. *Dollar spread/dollar depth* is the ratio of quoted spread to quoted depth, both expressed in dollars. $|Daily\ AR(1)|$ is the absolute value of first-order autocorrelation of daily returns. $|Variance\ ratio(5) - 1|$ is the absolute deviation of five-day variance ratio from one.

$\Delta Market\ cap$ and $\Delta Volatility$ are the post-issue equity market capitalization and standard deviation of daily return minus the corresponding pre-issue period measures. $\Delta Institutional\ holdings$ is the change in institutional holdings (by 13f institutions) divided by shares outstanding at issuing calendar year end from prior calendar year end. *Pre-issue price* is the average (log) price in pre-issue period. *NYSE* and *Public* are dummy variables. $\Delta Pre\ post$ is the number of days between the last day in pre-issue period and the first day in post-issue period. Estimates on the year-month dummies are not reported.

All change variables are expressed as deviation from control firms (issuer minus control). Control firms are matched based on size, book-to-market, and turnover before issuance, exchange, and industry. Heteroskedasticity-consistent industry-clustered *t*-statistics are in parentheses. *, **, and *** denote 10%, 5%, and 1% significance, respectively. The sample period is from July 1993 to May 2006.

	Liquidity measures						Efficiency measures											
	Δ Turnover	<i>t</i> -stat	Δ Trades	<i>t</i> -stat	Δ Amihud	<i>t</i> -stat	Δ OIBNUM	<i>t</i> -stat	Δ Spread	<i>t</i> -stat	Δ Depth	<i>t</i> -stat	Δ Spr/dep	<i>t</i> -stat	Δ AR(1)	<i>t</i> -stat	Δ VR(5) - 1	<i>t</i> -stat
Intercept	−0.144	(−1.00)	−0.214**	(−2.09)	−0.031	(−0.13)	0.048	(0.40)	−0.031	(−0.32)	0.505**	(2.22)	−0.171	(−0.93)	3.744	(0.44)	5.440	(0.94)
ΔSI	1.999**	(2.17)	2.062***	(3.44)	−2.364***	(−2.70)	−0.529	(−1.09)	0.120	(0.24)	1.348**	(2.40)	−1.207*	(−1.80)	−38.650	(−0.55)	−1.451	(−0.07)
$E(\Delta Volume)$	0.088***	(4.44)	0.036**	(2.08)	−0.060**	(−2.06)	0.006	(0.60)	0.022**	(2.05)	0.053	(1.37)	−0.039	(−1.44)	−0.670	(−0.40)	−0.183	(−0.32)
$\Delta Market\ cap$	0.195***	(5.63)	0.627***	(19.65)	−0.999***	(−22.37)	−0.190***	(−9.79)	−0.506***	(−12.26)	−0.718***	(−6.85)	−0.220***	(−4.30)	−6.026	(−0.63)	−1.693**	(−2.02)
$\Delta Volatility$	9.920***	(6.67)	9.493***	(6.75)	−0.272	(−0.28)	−2.304***	(−2.76)	2.907***	(2.65)	5.890***	(2.71)	0.001	(0.00)				
$\Delta Institutional\ holdings$	0.433***	(9.23)	0.156***	(2.91)	−0.465***	(−4.09)	−0.039	(−1.06)	−0.214***	(−3.77)	−0.119	(−0.97)	−0.111	(−1.52)	0.845	(0.10)	−0.803	(−0.56)
Pre-issue price $\times$ 100	4.064	(1.33)	5.769***	(2.68)	−1.823	(−0.65)	−1.626	(−1.45)	1.433	(0.81)	−7.365**	(−2.26)	−7.667	(−1.15)	84.538	(0.43)	−41.708	(−0.39)
NYSE	−0.063	(−1.63)	−0.112***	(−3.77)	−0.032	(−0.59)	0.049**	(2.11)	−0.012	(−0.64)	0.094	(0.98)	−0.013	(−0.26)	3.900	(1.02)	−0.683	(−0.56)
Public	−0.026	(−0.46)	0.058	(1.16)	0.098	(1.62)	−0.037	(−1.53)	0.059*	(1.87)	−0.094	(−0.99)	0.045	(0.64)	−16.492	(−1.11)	3.547	(1.43)
$\Delta Pre\ post \times 1000$	1.425*	(1.77)	0.313	(0.59)	−0.657	(−0.47)	−0.584	(−1.07)	−0.618	(−1.12)	−0.606	(−0.37)	2.892	(0.97)	−14.160	(−0.11)	−23.980	(−1.09)
Number of observations	846		846		846		846		846		846		846		846		846	
Adjusted R^2 (%)	38.25		56.60		58.13		33.45		58.20		36.19		20.12		10.47		59.58	

in short interest represent arbitrageurs covering short positions, and increases represent additional shorting. The increase in the average amount of shorting and covering that takes place following issuance is interpreted as dynamic hedging by convertible bond arbitrageurs.³⁵ The actual change in volume, $\Delta Volume$, is measured over the same period as *Dynamic_Hedge* and is also included in the regression.³⁶ This variable is defined as average (log) post-issue period volume minus average (log) pre-issue period volume. Table 9 presents results of the analysis in which the contemporaneous measures *Dynamic_Hedge* and $\Delta Volume$ replace Table 8's ΔSI and $E(\Delta Volume)$, respectively.³⁷ As with all other variables, these are presented as deviations from control firms. The evidence in Table 9 is similar to Table 8 and suggests a robust, positive role for convertible bond arbitrageurs in stock market liquidity.³⁸

5.7. Further evidence of convertible bond arbitrage: 2005–2006 Reg-SHO data

Ideally, the main analysis would measure short selling activity or the change in short interest during the few days surrounding issuance. However, the short-interest data are available only on a monthly basis and therefore do not perfectly capture short-sales transactions. We take advantage of newly available data on short selling activity (beginning in 2005, as a result of Regulation SHO) to investigate whether the increases in short interest that we observe using the monthly data are due to short-sales transactions close to the issue date. The SHO transactions data allow us to supplement the main analysis in two ways: (1) we are able to observe trading at the issue date and (2) we can examine changes in short selling activity following issuance. The latter is of interest since, if arbitrageurs dynamically hedge, we would expect to observe an increase in short selling following issuance.

Fig. 5 illustrates short selling activity near the convertible bond issue date for the sample of 64 issues for which Reg-SHO data are available (those in 2005 and 2006). The increase in short selling on Day 0 provides further evidence that we are identifying convertible bond arbitrage activity and not short selling due to other factors. The figure also suggests that the level of short selling activity following issuance is higher, which is

consistent with dynamic hedging by arbitrageurs. We explicitly test whether short selling increased.

Table 10 summarizes changes in short selling activity in stocks of convertible bond issuers between March 2005 and May 2006. The “pre-period” is defined as the 20 trading days ending one month prior to announcement. The “post-period” is defined as the 20 trading days beginning one month following issuance. The change is defined as the mean (or median) measure in post-period minus the mean (median) measure in pre-period.³⁹ For comparison, we also present results for control firms. A key observation from the table is consistent with dynamic hedging: short selling activity increases following issuance. Moreover, we do not find similar results for the control firms.

6. Conclusion

In this paper, we investigate the link between convertible bond arbitrage, liquidity, and stock prices with the goal of improving our understanding of the impact of arbitrageurs on market quality, measured by stock liquidity and efficiency. A typical convertible bond arbitrage strategy employs delta-neutral hedging in which an arbitrageur buys the convertible bond and sells the underlying equity at a specific delta. If the price of the stock increases, to keep the same delta hedge, the arbitrageur sells the stock. When the stock price decreases, the arbitrageur buys back the stock. Because aggregate trading demand tends to move in the opposite direction of this convertible bond arbitrage activity, arbitrageurs are expected to improve liquidity in the stock.

We examine changes in short interest near an event in which the convertible bond arbitrage strategy is widely used (bond issuance date), and are able to use aggregate data to estimate the equity positions taken by convertible bond arbitrageurs. Specifically, we examine the sensitivity of subsequent changes in liquidity and stock price-efficiency measures to this proxy for convertible bond arbitrage activity. We add to findings in previous studies that show average changes in equity market quality when new securities are introduced in that we are able to examine cross-sectional implications of the introduction of a particular trader type. This helps to shed additional empirical light on the issue of how the introduction of new securities that are used by arbitrageurs can impact overall market quality.

We show improvements in liquidity following issuance of convertible debt. More important, we find that the increase in liquidity is systematically related to our proxy for convertible bond arbitrage activity. This suggests positive liquidity spillovers due to the arbitrage activity in equity markets. We do not find evidence of a systematic relationship between arbitrage activity and stock return

³⁵ A second benefit of this measure relative to ΔSI is that both arbitrage activity and the liquidity and efficiency measures are calculated over the same window. We thank an anonymous referee for encouraging robustness analysis using synchronous time windows.

³⁶ Kumar, Sarin, and Shastri (1998) also include changes in log volume in their investigation of the impact of the introduction of options markets on stock market liquidity. In their regressions using volume, the authors focus on liquidity changes that are not volume-based (e.g., spread, depth, and the depth-to-spread ratio).

³⁷ Results are robust to defining $\Delta Volume$ as the change in buy volume, rather than the change in total volume.

³⁸ In addition, we used two alternative definitions for dynamic hedging: (1) the change in (log) standard deviation of monthly short interest and (2) change in (log) short interest during positive return months. The latter aims to capture the intuition that convertible bond arbitrageurs will short more when stock prices rise. Both measures yield results that are qualitatively similar to those presented in Table 9.

³⁹ Note that this “pre-” and “post-” period definition differs from that used in the main analysis. We tighten the window over which we measure transactions in order to maximize the number of issues for analysis, given that the SHO data do not begin until 2005.

Table 9

Regression of liquidity and efficiency changes on dynamic hedging activity and volume change.

This table presents results of a regression in which equity market liquidity and efficiency changes are regressed on the proxy for dynamic hedging activity (*Dynamic_Hedge*), change in volume (Δ *Volume*), and firm characteristics. *Dynamic_Hedge* is defined as the average (log) absolute value of short-interest changes during post-issue period minus the average (log) absolute value of short-interest changes during pre-issue period. The short-interest change is scaled by the number of shares outstanding in the prior month. Δ *Volume* is the post-issue average (log) daily volume in number of shares minus the pre-issue average (log) daily volume in number of shares. The “pre-issue period” is the six months ending one month prior to announcement. The “post-issue period” is the six months beginning one month after issuance.

Δ *Turnover*, Δ *Trades*, and Δ *Amihud* are changes in the post-issue log *Turnover*, log *Number of trades*, and log *Amihud*, respectively, from the pre-issue measures. Δ *OIBNUM*, Δ *Spread*, Δ *Depth*, and Δ *Spr/dep* are the percentage changes in the post-issue *OIBNUM*, *Percentage spread*, *Total depth/shares outstanding*, and *Dollar spread/dollar depth*, respectively, from the pre-issue measures. $\Delta|AR(1)|$ and $\Delta|VR(5) - 1|$ are the percentage changes in the post-issue $|Daily\ AR(1)|$ and $|Variance\ ratio(5) - 1|$, respectively, from the pre-issue measures.

Turnover is the average daily stock volume divided by shares outstanding. *Number of trades* is the average daily number of stock transactions. *Amihud* is the average daily ratio of absolute return to dollar volume (Amihud, 2002). *OIBNUM* is the average daily absolute difference between the numbers of buyer- and seller-initiated trades divided by their sum. *Percentage spread* is the difference between bid and ask quotes (time-weighted), expressed as percentage of bid–ask midpoint. *Total depth/shares outstanding* is the sum of bid and ask quoted depths divided by shares outstanding. *Dollar spread/dollar depth* is the ratio of quoted spread to quoted depth, both expressed in dollars. $|Daily\ AR(1)|$ is the absolute value of first-order autocorrelation of daily returns. $|Variance\ ratio(5) - 1|$ is the absolute deviation of five-day variance ratio from one.

Δ *Market cap* and Δ *Volatility* are the post-issue equity market capitalization and standard deviation of daily return minus the corresponding pre-issue period measures. Δ *Institutional holdings* is the change in institutional holdings (by 13f institutions) divided by shares outstanding at issuing calendar year end from prior calendar year end. *Pre-issue price* is the average (log) price in pre-issue period. *NYSE* and *Public* are dummy variables. Δ *Pre post* is the number of days between the last day in pre-issue period and the first day in post-issue period. Estimates on the year-month dummies are not reported.

All change variables are expressed as deviation from control firms (issuer minus control). Control firms are matched based on size, book-to-market, and turnover before issuance, exchange, and industry. Heteroskedasticity-consistent industry-clustered *t*-statistics are in parentheses. *, **, and *** denote 10%, 5%, and 1% significance, respectively. The sample period is from July 1993 to May 2006.

	Liquidity measures						Efficiency measures											
	Δ Turnover	<i>t</i> -stat	Δ Trades	<i>t</i> -stat	Δ Amihud	<i>t</i> -stat	Δ OIBNUM	<i>t</i> -stat	Δ Spread	<i>t</i> -stat	Δ Depth	<i>t</i> -stat	Δ Spr/dep	<i>t</i> -stat	Δ AR(1)	<i>t</i> -stat	Δ VR(5) - 1	<i>t</i> -stat
Intercept	0.151	(1.55)	-0.069	(-0.69)	-0.513**	(-2.13)	0.087	(0.87)	0.042	(0.51)	0.558***	(2.84)	-0.504	(-1.51)	0.817	(0.05)	4.798	(0.89)
Dynamic_Hedge	0.061***	(3.44)	0.005	(0.55)	-0.068***	(-3.16)	0.004	(0.48)	-0.027**	(-2.07)	0.065***	(2.67)	-0.035	(-1.46)	2.922	(0.78)	-0.244	(-0.41)
Δ Volume	0.774***	(24.57)	0.628***	(32.01)	-0.558***	(-13.26)	-0.116***	(-5.46)	-0.039	(-1.31)	0.309***	(4.88)	-0.367***	(-8.79)	6.330	(1.41)	-2.243*	(-1.79)
Δ Market cap	-0.163***	(-7.18)	0.347***	(13.79)	-0.761***	(-19.13)	-0.141***	(-6.75)	-0.490***	(-11.13)	-0.877***	(-6.02)	-0.063	(-1.36)	-8.749	(-0.90)	-0.812	(-1.00)
Δ Volatility	0.994*	(1.81)	2.618***	(5.97)	6.527***	(4.20)	-1.043	(-1.31)	3.593***	(2.91)	2.021	(1.11)	4.360***	(3.08)				
Δ Institutional holdings	0.242***	(6.12)	0.024	(0.45)	-0.324***	(-2.64)	-0.017	(-0.48)	-0.186***	(-3.05)	-0.223	(-1.56)	-0.022	(-0.27)	-2.492	(-0.36)	-0.221	(-0.14)
Pre-issue price $\times$ 100	-3.880***	(-3.28)	0.030	(0.02)	3.530	(1.35)	-1.177	(-1.18)	-0.163	(-0.08)	-11.195***	(-2.85)	-4.142	(-0.80)	136.899	(0.72)	-15.335	(-0.19)
NYSE	-0.002	(-0.08)	-0.059***	(-2.97)	-0.063	(-1.53)	0.037	(1.59)	-0.028	(-1.31)	0.120	(1.36)	-0.035	(-0.71)	-6.023	(1.39)	1.040	(-0.75)
Public	-0.031	(-1.41)	0.032	(1.39)	0.063	(1.47)	-0.029	(-1.15)	0.045*	(1.69)	-0.108	(-1.08)	0.024	(0.30)	-15.221	(-1.05)	3.542	(1.46)
Δ Pre post $\times$ 1000	0.405	(0.58)	0.208	(0.23)	4.431**	(2.14)	-1.085	(-1.60)	-0.182	(-0.21)	0.707	(0.31)	6.531	(0.97)	-54.640	(-0.19)	-22.680	(-0.45)
Number of observations	841		841		841		841		841		841		841		841		841	
Adjusted R^2 (%)	85.29		83.58		71.95		37.24		58.40		40.10		30.41		10.63		59.76	

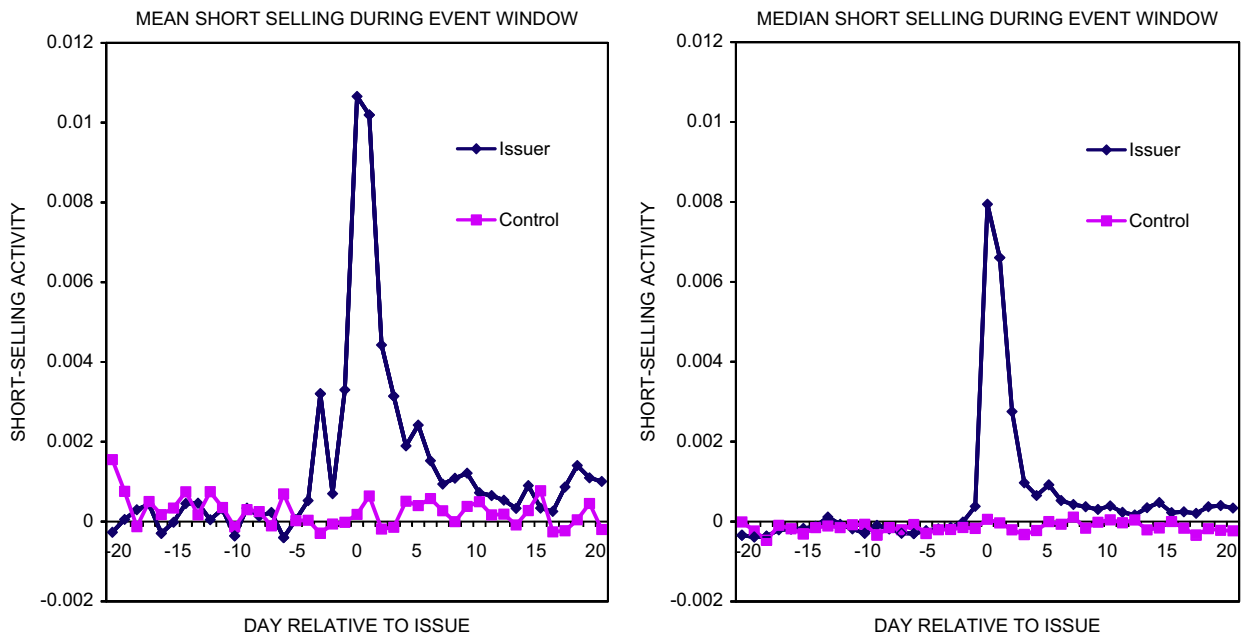


Fig. 5. Reg-SHO—short-selling activity for issuers and control firms (number of observations = 64). This chart shows short-selling activity during the event window (trading days –20 to +20) for convertible bond issuers and control firms between March 2005 and May 2006 (following the SEC’s enactment of Regulation SHO). Control firms are matched based on size, book-to-market, and turnover before issuance, exchange, and industry. The short-selling activity measure is the change from expected daily short-sales volume divided by total shares outstanding. Expected short-sales volume is the average short-sales volume from days –40 to –21 relative to issuance.

Table 10

Reg-SHO—changes in short-selling activity analysis.

This table shows the short-selling activity in stocks of convertible bond issuers and control firms between March 2005 and May 2006 (following the SEC’s enactment of Regulation SHO). Short-selling activity is measured using Reg-SHO data, which contain short-sale transactions data beginning January 2005. The “pre-period” is defined as the 20 trading days ending one month prior to announcement. The “post-period” is defined as the 20 trading days beginning one month following issuance. The change is defined as the mean measure in post-period minus the mean measure in pre-period. Control firms are matched based on size, book-to-market, and turnover before issuance, exchange, and industry.

Number of short sales is the daily number of stock transactions that involve short sales. *Short sales/shares outstanding* is the daily short-sales volume divided by the total shares outstanding. Industry- and time-clustered *t*-statistics of the changes are in parentheses. *, **, and *** denote 10%, 5%, and 1% significance, respectively.

Number of observations = 64.

	Issuing firm	Control firm
Pre-period number of short sales (log)	6.036	5.474
Post-period number of short sales (log)	6.295	5.623
Change in (log) number of short sales	0.258***	0.149**
<i>t</i> -stat	(4.68)	(2.57)
Pre-period short sales/shares outstanding (%)	0.298	0.227
Post-period short sales/shares outstanding (%)	0.345	0.230
Change in short sales/shares outstanding (%)	0.046*	0.003
<i>t</i> -stat	(1.79)	(0.10)

volatility and efficiency; however, we do find evidence of average changes in volatility measures near bond issuance. We perform a variety of robustness checks,

including controlling for the potential endogeneity of convertible bond arbitrage activity as well as other potential sources of volume. We find similar results.

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